Khulna University of Engineering & Technology
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Computer Graphics and Image Processing Laboratory
Course code: CSE 4102

Jungle Rest Stop

An interactive 3D scene built only from basic shapes
with vending machine, ATM, coffee machine & library



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Contents

1 Introduction **3**

2 The Scene in Pictures **3**

3 Tools and Environment **6**

4 System Design **6**
4.1 Modules . . . 6
4.2 Scene layout . . . 7

5 The Basic Shapes **7**
5.1 The vertices of each shape . . . 8

6 Transformations and Hierarchical Modelling **9**
6.1 The three basic transformations . . . 9
6.2 Order of operations . . . 10
6.3 Following vertices into the world . . . 10
6.4 The `Node` helper . . . 12
6.5 Example hierarchy: the human body . . . 13

7 Building Every Object from Basic Shapes **13**
7.1 Vending machine . . . 14
7.2 Products . . . 16
7.3 ATM . . . 17
7.4 Coffee machine, TV and aquarium . . . 17
7.5 Furniture and lamps . . . 20
7.6 People . . . 25
7.7 Cars . . . 27
7.8 The building . . . 29
7.9 The jungle . . . 30

8 Lighting and Shading **34**
8.1 Illumination model . . . 34
8.2 Shiny materials: polished metal and glossy paint . . . 35
8.3 Light types and where they are shown . . . 36
8.4 Shading: where the equation is evaluated . . . 36
8.5 Choosing lights every frame . . . 37

9 Day and Night **38**

10 Textures **38**
10.1 Procedural textures . . . 38
10.2 How textures are applied . . . 39

11 Shadows **40**

12 Sky, Fog, Tone Mapping and Wind **41**

13 Basic Mode versus Enhanced Mode **41**

14 Interaction: Using the Machines **42**
14.1 Look-and-press picking . . . 42
14.2 ATM: step by step . . . 43
14.3 Coffee machine and TV . . . 44

15 NPC Visitors **44**
15.1 Life of a visitor . . . 44
15.2 Driving and parking . . . 44
15.3 Walking and activities . . . 45

16 AI, Particles, Text and Other Effects **47**
16.1 All the AI in one table . . . 47
16.2 Particles . . . 47
16.3 Text rendering . . . 48
16.4 Collision detection . . . 48
16.5 Sound . . . 48
16.6 Fog, frame cap and timing . . . 49

17 The Player and the Cameras **49**

18 Performance on a Weak GPU **50**
18.1 Techniques . . . 50
18.2 Measured speed . . . 51

19 Conclusion **51**

1 Introduction

Jungle Rest Stop is a real-time 3D scene made with OpenGL 3.3. A small rest stop stands in a clearing of a thick jungle, next to a parking lot and a road. The main rule of the course is followed everywhere: **every object is built only from basic shapes** (cube, cylinder, sphere, cone, torus, wedge and a few more) placed with translation, rotation and scaling. No 3D model files and no image files are loaded; all shapes and all textures are made by the program.

2 The Scene in Pictures

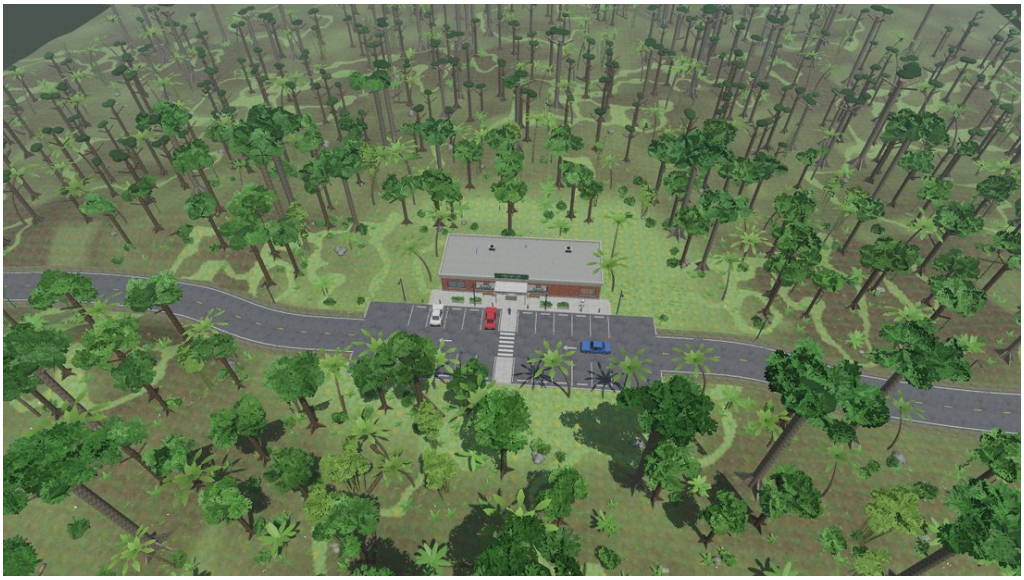


Figure 1: The rest stop in its jungle clearing.



Figure 2: The rest stop by day: building, sidewalk, parking lot and street lamps.



Figure 3: The main lounge: sofas, TV wall, coffee counter and the three vending machines.



Figure 4: The three working vending machines: cold drinks, crispy snacks and sweet treats.



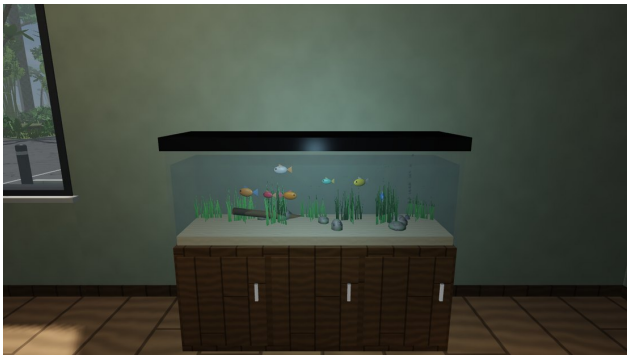
(a) ATM



(b) Coffee machine (brewing)



(c) Library



(d) Aquarium in the library

Figure 5: The ATM room, the coffee counter and the library.



(a) Sunrise 6:18



(b) Day 10:30



(c) Sunset 17:36



(d) Night 21:00

Figure 6: The same view at four times of day.

3 Tools and Environment

Item	Used
Language	C++17 (MinGW-w64 / MSYS2 `g++`)
Graphics API	OpenGL 3.3 Core profile, GLSL 330
Window / input	GLFW 3
OpenGL loader	GLAD 2
Text	Windows GDI renders one font atlas at start-up (Consolas)
Sound	Windows `waveOut` (winmm); 14 short CC0 (public domain) recordings
Assets	**no models, no images**: all meshes and textures are generated in code; only the sounds are
Test PC	Intel Core i5-1135G7, **Intel Iris Xe (integrated GPU)**, 1280×720 window, 4×MSAA

4 System Design

4.1 Modules

The code is split into small modules. Each one has a single job (Figure 7).

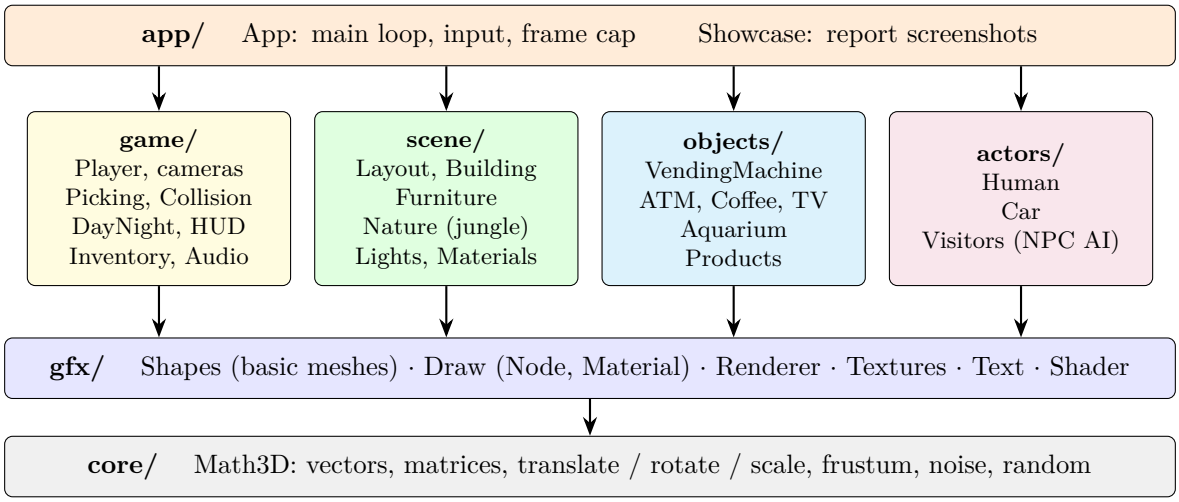


Figure 7: Module overview. Arrows mean "uses".

4.2 Scene layout

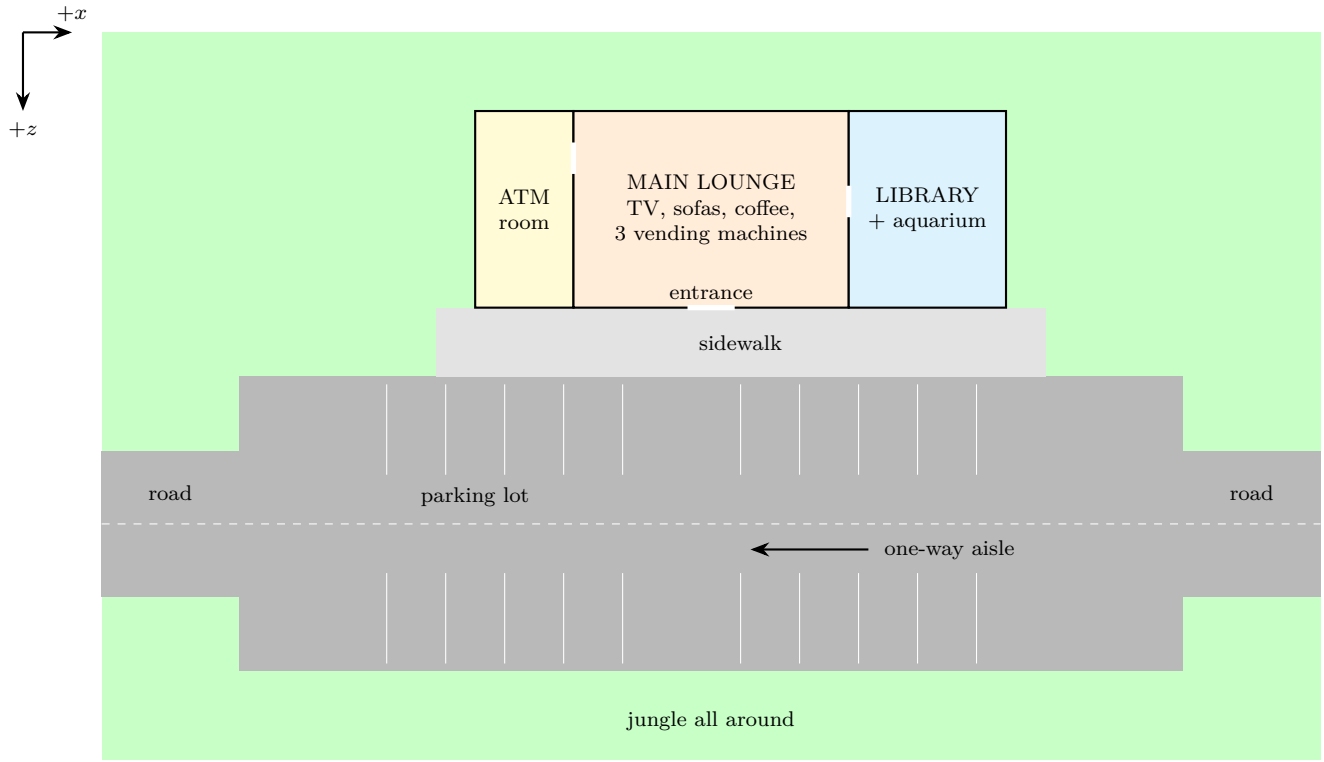


Figure 8: Top view of the layout (1 grid unit = 1 m). The building faces $+z$, towards the parking lot.

5 The Basic Shapes

Everything is built from the small set of *unit* meshes in Table 1. Each one is centred at the origin with a size of about one unit. They are created once at start-up (`gfx/Shapes.cpp`) and kept on the GPU.

Table 1: Basic shapes and how they are generated

Shape	How it is generated	Typical use
Cube	6 faces × 4 vertices (flat normals), $[-0.5, 0.5]^3$	walls, panels, keys, books
Cylinder	ring of 24 vertices at $y = \pm 0.5$, radius 0.5, plus two caps	legs, poles, cans, wheels
Cylinder (low)	same, 8 sides only	tree trunks, bamboo
Sphere	latitude/longitude grid, 14 × 20	heads, LEDs, bulbs
Sphere (low)	5 × 8 grid	foliage blobs, rocks
Cone	cylinder with top radius 0	fish tails, snake-plant leaves
Truncated cone	cylinder with top radius 0.35	cups, pots, lamp shades, poles
Torus	ring radius 0.4, tube radius 0.1	clock rim, tyres on rings, collars
Hemisphere	top half of a sphere + bottom disc	hair, domes, bowl
Wedge	right-angled triangle extruded along x	bonnet, buttress roots, fins
Tapered box	bottom 1 × 1, top 0.7 × 0.6	car cabin, chest, lamp head
Quad	one square in the xy plane	palm fronds, vines
Cross card	two crossed quads, $y \in [0, 1]$	grass, ferns, flowers
Leaf card	three crossed quads	leaf clusters of trees
Coil	tube swept along a helix (5 turns)	vending machine spirals
Terrain grid	150 × 150 grid displaced by $h(x, z)$	the ground

The curved shapes come from simple parametric formulas. For a cylinder (radius r, angle $\theta_i = 2\pi i/n$):

$$\mathbf{p}_i^{top} = (r\cos\theta_i,\ 0.5,\ r\sin\theta_i), \qquad \mathbf{p}_i^{bottom} = (r\cos\theta_i,\ -0.5,\ r\sin\theta_i), \qquad \mathbf{n}_i = (\cos\theta_i,\ 0,\ \sin\theta_i).$$

A sphere uses $\mathbf{p}(\phi,\theta) = 0.5\,(\sin\phi\cos\theta,\ \cos\phi,\ \sin\phi\sin\theta)$ with normal $\mathbf{n} = 2\mathbf{p}$. A torus uses $\mathbf{p}(u,w) = (R{+}r\cos w)(\cos u, 0, \sin u){+}(0, r\sin w, 0)$. The coil follows the helix $\mathbf{c}(t) = (R\cos 2\pi kt,\ R\sin 2\pi kt,\ L(t{-}0.5))$, and a small circle is placed around each point of it.

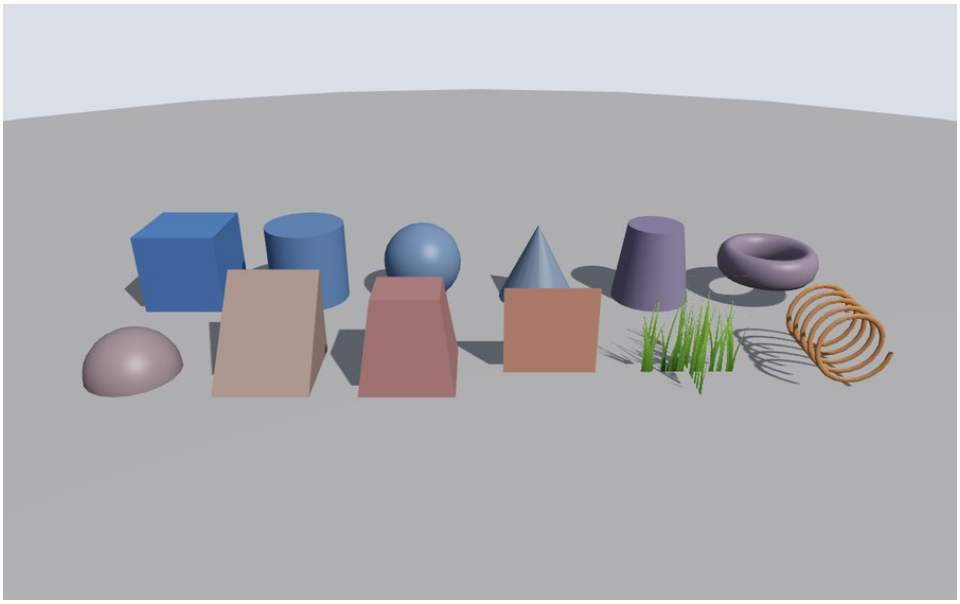
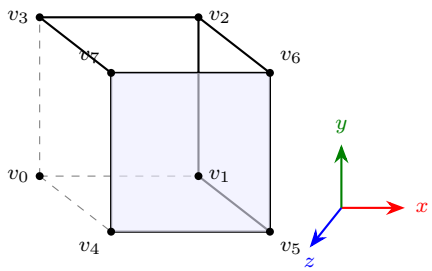


Figure 9: The basic shapes rendered by the program (cube, cylinder, sphere, cone, truncated cone, torus, hemisphere, wedge, tapered box, quad, cross card, coil).

5.1 The vertices of each shape

This part lists the exact vertices that are stored on the GPU. Every object in the scene is only a copy of one of these lists, moved by a matrix (Section 6.3).

Flat shapes (corners are written out). The cube is the most used shape. It has 8 corners (Figure 10), but every face gets its own 4 vertices so that each face has one flat normal. So the cube stores $6 \times 4 = 24$ vertices and $6 \times 2 = 12$ triangles. Each face (a, b, c, d) is split into the triangles (a, b, c) and (a, c, d).



Corner	(x, y, z)	Corner	(x, y, z)
v_0	$(-.5, -.5, -.5)$	v_4	$(-.5, -.5, .5)$
v_1	$(.5, -.5, -.5)$	v_5	$(.5, -.5, .5)$
v_2	$(.5, .5, -.5)$	v_6	$(.5, .5, .5)$
v_3	$(-.5, .5, -.5)$	v_7	$(-.5, .5, .5)$

Faces: front $(v_4v_5v_6v_7)$, back $(v_1v_0v_3v_2)$, top $(v_7v_6v_2v_3)$, bottom $(v_0v_1v_5v_4)$, right $(v_5v_1v_2v_6)$, left $(v_0v_4v_7v_3)$.

Figure 10: The unit cube: 8 corners, 6 faces. The front face (shaded) looks towards $+z$.

The other flat shapes are built in the same way. Their corners are:

Table 2: Corner vertices of the flat shapes

Shape	Corner vertices	Vertices / triangles
Wedge	left side $w_0(-.5,-.5,.5)$, $w_1(-.5,-.5,-.5)$, $w_2(-.5,.5,-.5)$; right side w_3, w_4, w_5 = the same with $x = +.5$. Faces: 2 triangles $(w_0w_2w_1)$, $(w_3w_4w_5)$; bottom, back and sloping quads	18 / 8
Tapered box	bottom $b_0(-.5,-.5,.5)$, $b_1(.5,-.5,.5)$, $b_2(.5,-.5,-.5)$, $b_3(-.5,-.5,-.5)$; top $t_0(-.35,.5,.3)$, $t_1(.35,.5,.3)$, $t_2(.35,.5,-.3)$, $t_3(-.35,.5,-.3)$. Side normals: $\mathbf{n} = \text{normalize}((b-a)\times(d-a))$	24 / 12
Quad	$(-.5,-.5,0)$, $(.5,-.5,0)$, $(.5,.5,0)$, $(-.5,.5,0)$, normal $(0,0,1)$	4 / 2
Cross card	quad 1: $(-.5,0,0)$, $(.5,0,0)$, $(.5,1,0)$, $(-.5,1,0)$; quad 2: $(0,0,-.5)$, $(0,0,.5)$, $(0,1,.5)$, $(0,1,-.5)$. The base is at $y = 0$, so a scaled card stands on the ground	8 / 4
Leaf card	three quads of size 1×1 through the centre, one in each plane (xy, yz, xz)	12 / 6

Round shapes (vertices come from a formula). A round shape has too many vertices to write by hand, so they come from a loop over an angle. The cylinder, cone and truncated cone use the same loop with n steps, $\theta_i = 2\pi i/n$ for $i = 0\ldots n$, bottom radius $r_b = 0.5$ and top radius r_t:

$$\mathbf{p}_i^{top} = (r_t\cos\theta_i,\ 0.5,\ r_t\sin\theta_i), \qquad \mathbf{p}_i^{bot} = (0.5\cos\theta_i,\ -0.5,\ 0.5\sin\theta_i).$$

Each pair of neighbours makes one side quad $(\mathbf{p}_i^{bot}, \mathbf{p}_{i+1}^{bot}, \mathbf{p}_{i+1}^{top}, \mathbf{p}_i^{top})$. A cap is a centre vertex $(0, \pm0.5, 0)$ plus the ring, joined as a fan of triangles. For example, the first vertices of the 24-sided cylinder are $\mathbf{p}_0^{top} = (0.5, 0.5, 0)$, $\mathbf{p}_1^{top} = (0.483, 0.5, 0.129)$, $\mathbf{p}_2^{top} = (0.433, 0.5, 0.25)$ and $\mathbf{p}_6^{top} = (0, 0.5, 0.5)$.

Table 3: Vertex formulas and counts of the round shapes

Shape	Vertex formula	Vertices / triangles
Cylinder	loop above, $r_t = 0.5$, $n = 24$, two caps	102 / 96
Cylinder (low)	same, $n = 8$	38 / 32
Cone	$r_t = 0$ (all top vertices meet at the tip $(0, 0.5, 0)$), $n = 20$, bottom cap only	64 / 60
Truncated cone	$r_t = 0.35$, $n = 20$, two caps	86 / 80
Sphere	$\mathbf{p}_{ij} = 0.5(\sin\phi_i\cos\theta_j,\ \cos\phi_i,\ \sin\phi_i\sin\theta_j)$, $\phi_i = \pi i/14$, $\theta_j = 2\pi j/20$	315 / 560
Sphere (low)	same with 5 rings and 8 sectors	54 / 80
Hemisphere	same with $\phi \in [0, \pi/2]$ (8 rings) + a flat disc at $y = 0$	211 / 340
Torus	$\mathbf{p}_{ij} = (0.4 + 0.1\cos w_j)(\cos u_i, 0, \sin u_i) + (0, 0.1\sin w_j, 0)$, 28×10 steps	319 / 560
Coil	a 6-sided circle around each of 101 points of the helix $\mathbf{c}(t)$	606 / 1200

6 Transformations and Hierarchical Modelling

6.1 The three basic transformations

All matrices are 4×4 and column-major, so they can go straight to OpenGL.

$$T(t) = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad S(s) = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

R_x and R_z work in the same way. Rotation about any axis $\mathbf{a}$ uses Rodrigues' formula $R = \cos\theta\, I + (1-\cos\theta)\,\mathbf{a}\mathbf{a}^T + \sin\theta\,[\mathbf{a}]_\times$. The camera uses the usual `lookAt` view matrix and a perspective projection

$$P = \begin{bmatrix} \frac{1}{a\tan(f/2)} & 0 & 0 & 0 \\ 0 & \frac{1}{\tan(f/2)} & 0 & 0 \\ 0 & 0 & -\frac{z_f+z_n}{z_f-z_n} & -\frac{2z_f z_n}{z_f-z_n} \\ 0 & 0 & -1 & 0 \end{bmatrix},$$

where f is the field of view, which the user changes with the mouse wheel (see Section 17).

6.2 Order of operations

Every part of every object is placed with

$$M_{part} = M_{parent} \cdot T(\text{position}) \cdot R(\text{rotation}) \cdot S(\text{size}).$$

The unit shape is first **scaled** to the right size, then **rotated**, then **moved** to its place inside the parent, and finally the parent's own matrix carries it into the world. Normals use the normal matrix $N = (M^{-1})^T$ (upper 3×3 part), which keeps lighting correct even when a shape is stretched.

6.3 Following vertices into the world

The objects fall into four groups. Objects in the same group are built in the same way; only the numbers change. For each group, one part is worked out by hand below: its unit vertices (Section 5.1) are pushed through S, then R, then T, then the parent matrices.

Table 4: The four groups of objects

Group	Objects	Main shapes	Transformations used
A. Box-built	vending machines, ATM, counters, sofa, bookshelf, tables, walls, doors, TV	cube	S and T only (a few small tilts)
B. Round	wheels, cans, cups, lamps, poles, coffee machine parts, clock	cylinder, cone, sphere, torus	S, T and a 90° turn to lay the shape on its side
C. Jointed	the man, the woman, the car (door, wheels)	all of them	chains of $T \cdot R$ nodes; only the angles change when they move
D. Plants	trees, palms, bamboo, ferns, grass	low cylinder, wedge, cards	R_y around the trunk, random yaw and size

A. Box-built objects: the side panel of a vending machine. The first vending machine stands at $(2.4, 0, -4.5)$ and is not turned. Its right side panel is a cube with $S = (0.04, 1.86, 0.8)$ and $T = (0.48, 1.0, 0)$ inside the machine:

$$\mathbf{p}_{world} = T(2.4, 0, -4.5) \cdot T(0.48, 1.0, 0) \cdot S(0.04, 1.86, 0.8) \cdot \mathbf{v}.$$

For corner $v_6 = (0.5, 0.5, 0.5)$: scaling gives $(0.02, 0.93, 0.40)$, moving inside the machine gives $(0.50, 1.93, 0.40)$, and moving the machine into the room gives $(2.90, 1.93, -4.10)$. All 8 corners:

Table 5: The 8 corners of the vending machine side panel, step by step

Corner	unit $\mathbf{v}$	after S	after T (machine space)	world
v_0	$(-.5, -.5, -.5)$	$(-.02, -.93, -.40)$	$(.46, .07, -.40)$	$(2.86, 0.07, -4.90)$
v_1	$(\ .5, -.5, -.5)$	$(\ .02, -.93, -.40)$	$(.50, .07, -.40)$	$(2.90, 0.07, -4.90)$
v_2	$(\ .5,\ .5, -.5)$	$(\ .02,\ .93, -.40)$	$(.50, 1.93, -.40)$	$(2.90, 1.93, -4.90)$
v_3	$(-.5,\ .5, -.5)$	$(-.02,\ .93, -.40)$	$(.46, 1.93, -.40)$	$(2.86, 1.93, -4.90)$
v_4	$(-.5, -.5,\ .5)$	$(-.02, -.93,\ .40)$	$(.46, .07,\ .40)$	$(2.86, 0.07, -4.10)$
v_5	$(\ .5, -.5,\ .5)$	$(\ .02, -.93,\ .40)$	$(.50, .07,\ .40)$	$(2.90, 0.07, -4.10)$
v_6	$(\ .5,\ .5,\ .5)$	$(\ .02,\ .93,\ .40)$	$(.50, 1.93,\ .40)$	$(2.90, 1.93, -4.10)$
v_7	$(-.5,\ .5,\ .5)$	$(-.02,\ .93,\ .40)$	$(.46, 1.93,\ .40)$	$(2.86, 1.93, -4.10)$

The panel is 4 cm thick ($2.86 \to 2.90$), 1.86 m tall ($0.07 \to 1.93$) and 80 cm deep ($-4.90 \to -4.10$), exactly as asked. The other two machines only change the first T to $x = 3.7$ and $x = 5.0$. The ATM, the sofa, the shelves and the walls are made of many such boxes, each with its own S and T (see the part tables in Section 7).

B. Round objects: a car tyre. A tyre is a cylinder of size $S = (0.64, 0.22, 0.64)$ turned by $R_z(90°)$, so its axis points sideways along x. It sits in the wheel node $T(-0.78, 0.32, 1.325)$ (front wheel, driver side) of the car. With $R_z(90°)$: $(x, y, z) \mapsto (-y,\ x,\ z)$.

Table 6: Three vertices of the front tyre (car space)

Vertex	unit $\mathbf{p}$	after S	after $R_z(90°)$	after wheel T
$\mathbf{p}_0^{top}$	$(0.5, 0.5, 0)$	$(0.32, 0.11, 0)$	$(-0.11, 0.32, 0)$	$(-0.89, 0.64, 1.325)$
$\mathbf{p}_0^{bot}$	$(0.5, -0.5, 0)$	$(0.32, -0.11, 0)$	$(0.11, 0.32, 0)$	$(-0.67, 0.64, 1.325)$
$\mathbf{p}_6^{top}$	$(0, 0.5, 0.5)$	$(0, 0.11, 0.32)$	$(-0.11, 0, 0.32)$	$(-0.89, 0.32, 1.645)$

The top of the tyre is at $y = 0.64$, which is twice the wheel radius, so the tyre touches the ground at $y = 0$. The outer face is at $x = -0.89$ and the inner face at $x = -0.67$ (22 cm wide). When the car moves, two more matrices are put in front: the spin $R_x(\theta_{spin})$ and the car's own $T(\mathbf{p}_{car})R_y(\psi)$. A can, a cup or a lamp pole uses the same cylinder vertices, just standing up (no R_z).

C. Jointed models: the knee of a person and the car door. A person is a chain of nodes (Figure 11). The knee of the man's left leg is found with

$$\mathbf{p}_{knee} = \underbrace{T(\mathbf{p})R_y(\psi)}_{\text{person}} \cdot \underbrace{T(0, 0.93, 0)}_{\text{pelvis}} \cdot \underbrace{T(0.092, -0.03, 0)\, R_x(\theta_{thigh})}_{\text{hip}} \cdot \underbrace{T(0, -0.45, 0)}_{\text{knee}} \cdot (0, 0, 0).$$

$R_x(\theta)$ maps $(x, y, z) \mapsto (x,\ y\cos\theta - z\sin\theta,\ y\sin\theta + z\cos\theta)$. In person space:

- standing, $\theta_{thigh} = 0$: knee at $(0.092,\ 0.90 - 0.45,\ 0) = (0.092,\ 0.45,\ 0)$;
- walking, leg forward, $\theta_{thigh} = -26°$: $(0.092,\ 0.90 - 0.45\cos 26°,\ 0.45\sin 26°) = (0.092,\ 0.496,\ 0.197)$.

The knee moves 20 cm forward and 4.6 cm up, and the shin and the foot follow it because they are its children. The thigh itself is a truncated cone flipped by $R_x(180°)$, which maps $(x, y, z) \mapsto (x, -y, -z)$: the wide bottom ring vertex $(0.5, -0.5, 0)$ becomes, after $S(0.17, 0.45, 0.17)$, $R_x(180°)$ and $T(0, -0.225, 0)$, the point $(0.085, 0, 0)$, which is at the hip. So the wide end of the thigh is at the top, like a real leg.

The woman uses exactly the same chain; her model only has $S(0.95)$ at the root and a width factor 0.94, so her hip is at $x = 0.092 \times 0.94 = 0.086$.

The **car door** works the same way with R_y. Its hinge is at $\mathbf{h} = (-0.905, 0, 0.73)$ in car space, and the outer door panel is a cube with $S = (0.04, 0.534, 1.0)$ and $T = (-0.01, 0.582, -0.5)$ inside the

door node $T(\mathbf{h})R_y(\theta_{door})$. The rear-bottom corner $v_0 = (-.5, -.5, -.5)$ becomes $(-0.03, 0.315, -1.0)$ in door space. Then, with $R_y(\theta)$: $x' = x\cos\theta + z\sin\theta$, $z' = -x\sin\theta + z\cos\theta$:

- door closed, $\theta = 0°$: $(-0.03, 0.315, -1.0) + \mathbf{h} = (-0.935,\ 0.315,\ -0.27)$;
- door open, $\theta = 70°$: $(-0.950,\ 0.315,\ -0.314) + \mathbf{h} = (-1.855,\ 0.315,\ 0.416)$.

The back edge of the door swings 92 cm out to the side and moves forward, while the points on the hinge line do not move at all.

Windows: a matrix made straight from the corner points. Car windows are not rectangles, so they are not made with $T \cdot R \cdot S$. Instead the matrix is built from the corner points the window must have. For a parallelogram with corners $\mathbf{a}$, $\mathbf{b}$, $\mathbf{d}$ (and $\mathbf{b} + \mathbf{d} - \mathbf{a}$):

$$M = \begin{bmatrix} \mathbf{b}-\mathbf{a} & \mathbf{d}-\mathbf{a} & t\,\hat{\mathbf{n}} & \mathbf{a} + \frac{1}{2}(\mathbf{b}-\mathbf{a}) + \frac{1}{2}(\mathbf{d}-\mathbf{a}) \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \hat{\mathbf{n}} = \frac{(\mathbf{b}-\mathbf{a}) \times (\mathbf{d}-\mathbf{a})}{|(\mathbf{b}-\mathbf{a}) \times (\mathbf{d}-\mathbf{a})|}$$

where t is the glass thickness. Then the cube corner $(-.5, -.5, 0)$ lands exactly on $\mathbf{a}$, $(.5, -.5, 0)$ on $\mathbf{b}$ and $(-.5, .5, 0)$ on $\mathbf{d}$. A triangle is done the same way with the wedge: $w_1 \to \mathbf{p}_0$, $w_0 \to \mathbf{p}_1$, $w_2 \to \mathbf{p}_2$. Each side window is one parallelogram plus one triangle (a small quarter glass at the front, or a painted panel at the back), so no glass sticks out past the pillars.

D. Plants: the root fins of a rainforest tree. A tree has 4 buttress fins around its trunk. Fin i is a wedge with $S = (0.13, 1.5, 1.2)$ and $T = (0, 0.75, 0.6)$ inside a node turned by $R_y(90°i)$ (plus a small random angle):

- bottom-front vertex $w_0 = (-.5, -.5, .5)$: $S \to (-0.065, -0.75, 0.6)$, $T \to (-0.065, 0, 1.2)$, then $R_y(90°) \to (1.2,\ 0,\ 0.065)$;
- top-back vertex $w_2 = (-.5, .5, -.5)$: $S \to (-0.065, 0.75, -0.6)$, $T \to (-0.065, 1.5, 0)$, then $R_y(90°) \to (0,\ 1.5,\ 0.065)$.

So each fin starts on the trunk 1.5 m above the ground and reaches the ground 1.2 m away from it, and the four R_y angles spread the fins around the tree. Grass and ferns are cross cards placed with $T(\mathbf{p})R_y(\text{random})S(w, h, w)$: the card vertex $(0.5, 1, 0)$ becomes the tip of a blade at height h.

6.4 The `Node` helper

Objects are written as trees of `Node`s. A child node only stores the extra transform relative to its parent, so moving the parent moves all its children too.

```
struct Node {
    DrawSink* sink; Mat4 M; // M = this node's world matrix
    Node at(const Vec3& p, const Vec3& rotDeg) const // child: M * T(p) * R(rot)
        { return Node(*sink, rotateYXZ(translate(M, p), rotDeg)); }
    Node rotX(float deg) const { return rot(deg, Vec3(1,0,0)); }
    void part(MeshId mesh, Vec3 pos, Vec3 size, const Material& m, Vec3 rotDeg = {}) const
        { sink->add(mesh, scale(rotateYXZ(translate(M, pos), rotDeg), size), m); }
};
// Example: a sofa arm = a box plus a "rolled" cylinder lying along Z
n.box({x, 0.42f, 0.0f}, {0.16f, 0.44f, 0.88f}, fabric);
n.part(M_CYLINDER, {x, 0.64f, 0.0f}, {0.19f, 0.88f, 0.19f}, fabric, {90, 0, 0});
```

The same object code can send its shapes to different "sinks": the static world, the per-frame list, a counter (for the tables in this report), or a collector that keeps every part so it can be drawn alone (for the parts catalogues).

6.5 Example hierarchy: the human body

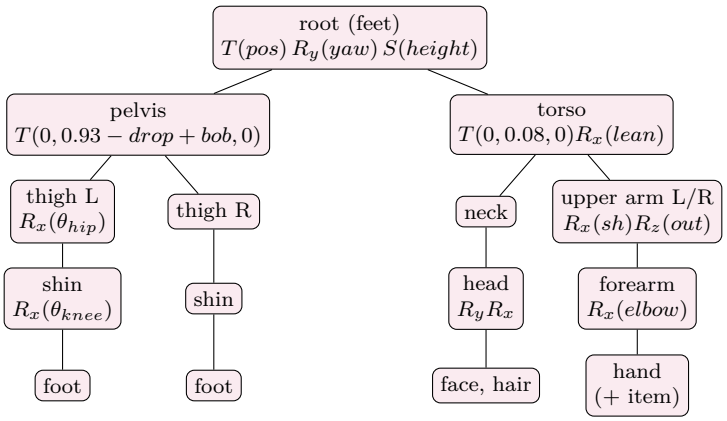


Figure 11: Scene graph of a person. The pose only changes the joint angles θ, while every body part stays attached to its parent.

7 Building Every Object from Basic Shapes

Table 7 lists how many basic shapes each object uses and which ones. The numbers are counted by the program itself (a counting sink). After the table, every object is shown in the same way:

- a picture with the **whole object** on the left and a **parts catalogue** on the right. The catalogue shows every different basic shape of the object *alone* in its own numbered tile, with the shape name, how many times it is used ($\times 4$ = four identical copies) and its size (s_x, s_y, s_z), which is exactly the scale S used in its matrix. Each tile keeps the part's rotation and is zoomed so the part fills the tile, so small parts (keys, eyes) are as easy to see as large ones; the size label tells the real size. Very big objects show their 36 biggest parts;
- a short table of the parts, grouped by shape and texture: how many, their colours (small squares), the texture (a small sample of it) and the size of the largest part. This table is also written by the program, so it always matches the code.

For the most important objects, a second table gives the names, sizes and transformations of the main parts.

Table 7: Number of basic shapes per object (Cyl-L / Sph-L = low-poly versions, TCone = truncated cone, TBox = tapered box)

Object	Total	Cube	Cyl	Cyl-L	Sph	Sph-L	Cone	TCone	Torus	Hemi	Wedge	TBox	Quad	Cross	Leaf	Coil
Vending machine	**287**	79	177	–	3	–	–	16	–	–	–	–	–	–	–	12
ATM	**49**	48	–	–	–	–	–	–	–	1	–	–	–	–	–	–
Coffee machine	**32**	16	9	–	1	–	–	2	4	–	–	–	–	–	–	–
Television (frame)	**5**	4	–	–	1	–	–	–	–	–	–	–	–	–	–	–
Sofa (3 seats)	**21**	12	9	–	–	–	–	–	–	–	–	–	–	–	–	–
Armchair	**13**	6	7	–	–	–	–	–	–	–	–	–	–	–	–	–
Bookshelf	**31**	21	–	–	–	–	–	–	–	–	–	–	10	–	–	–
Woman (standing)	**50**	7	6	–	24	–	–	10	1	1	–	1	–	–	–	–
Man (walking)	**49**	8	6	–	24	–	–	8	1	1	–	1	–	–	–	–
Person + chair	**56**	15	6	–	24	–	–	8	1	1	–	1	–	–	–	–
Car + driver (any colour)	**144**	76	23	–	24	–	–	8	2	1	9	1	–	–	–	–
Rainforest tree	**30**	–	–	5	–	3	–	–	–	–	4	–	–	–	18	–
Coconut palm	**37**	–	–	14	–	5	–	–	–	–	–	–	18	–	–	–
Bamboo clump	**130**	–	–	100	–	–	–	–	–	–	–	–	–	–	30	–
Banana plant	**28**	–	–	6	–	22	–	–	–	–	–	–	–	–	–	–
Street lamp	**5**	1	2	–	–	–	–	1	–	–	–	1	–	–	–	–
Pendant lamp	**5**	–	3	–	1	–	–	1	–	–	–	–	–	–	–	–
Aquarium	**76**	12	–	1	35	6	7	1	–	–	7	–	–	7	–	–
6 products	**23**	9	12	–	1	–	–	1	–	–	–	–	–	–	–	–
Table set	**23**	10	2	–	3	1	–	1	–	1	–	–	–	–	5	–
Wall clock	**17**	14	2	–	–	–	–	–	1	–	–	–	–	–	–	–
Bush+fern+rock+log	**23**	–	–	4	–	5	–	–	–	3	–	–	5	1	5	–

7.1 Vending machine

The machine is $1.0 \times 1.95 \times 0.8$ m. Its origin is at the floor in the middle, and it faces $+z$. Its *cabinet* is drawn once into the static world. The moving parts (spirals, products, keys, LEDs, flap and coins) are drawn every frame.



Figure 12: Vending machine: the whole object and its parts.

Table 8: Vending machine: the 287 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cylinder	177	+11 (glows)	–	$0.064 \times 0.11 \times 0.064$
cube	60	+8 (see-through, glows)	–	$0.64 \times 1.14 \times 0.02$
cube	19		Brushed metal	$1 \times 1.86 \times 0.04$
truncated cone	16		–	$0.062 \times 0.04 \times 0.062$
coil	12		–	$1 \times 1 \times 1$
sphere	3	(glows)	–	$0.022 \times 0.022 \times 0.022$

Table 9: Main parts of the vending machine (sizes and positions in metres, in the machine's own space)

Part	Shape	Size (s_x, s_y, s_z)	Position / rotation
Back, sides, top, base	cube	back $(1, 1.86, .04)$, side $(.04, 1.86, .8)$	back $(0, 1.0, -.38)$, sides $(\pm.48, 1.0, 0)$, top $(0, 1.93, 0)$
4 levelling feet	cylinder	$(.06, .05, .06)$	$(\pm.42, .025, \pm.32)$
Lit marquee (title)	cube, emissive	$(.94, .12, .02)$	$(0, 1.86, .39)$
Display cavity + tube light	cube, cylinder	tube $(.022, .6, .022)$	tube rotated $R_z(90°)$ at $(-.15, 1.76, .30)$
4 shelves A–D	cube	$(.62, .015, .38)$	$y_r = 1.555,\ 1.285,\ 1.015,\ 0.745$
12 spirals	coil	radius .055, 5 turns, .30 long	$(x_c, y_r + .065, -.06)$, $R_z(-spin)$
48 products	see Table 10		4 per slot, $z = .0525 - .075k$
Front glass	cube, $\alpha = 0.18$	$(.62, 1.14, .008)$	$(-.15, 1.22, .385)$
Control panel + LCD	cube	$(.27, 1.30, .04)$	$(.33, 1.20, .38)$
9 keys (A B C D 1 2 3 CLR OK)	cube	$(.05, .045, .014)$	grid $x = .255/.325/.395$, $y = 1.50 - .06\,row$; pushed in by .007 m when pressed
3 status LEDs	sphere, emissive	.022	ready (green), vending (amber), item (blue)
Coin slot, note slot, card pad, coin cup	cube	small plates	$y = 1.23,\ 1.12,\ 1.00,\ 0.86$
Coin-return lever	cube	$(.02, .05, .02)$	$R_x(40° \cdot press)$
Collection flap	cube (child node)	$(.58, .34, .015)$	hinge node $T(-.15, .57, .39) \cdot R_x(\theta_{flap})$
Returned coins	cylinder	$(.024, .004, .024)$	fall with gravity into the cup

How it works. Each machine is a small *state machine* (Figure 13). Money is kept as an integer number of Taka, so there are no rounding errors. When you press **OK**, the spiral turns 360° in 1.6 s, and the products move forward by one place (0.075 m). The front product then becomes a free body with velocity $\mathbf{v}$, gravity $g = 9.81\,\mathrm{m/s^2}$ and spin. When it hits the tray floor it bounces with restitution 0.32:

$$\mathbf{v}_y \leftarrow \mathbf{v}_y - g\,\Delta t, \qquad \mathbf{p} \leftarrow \mathbf{p} + \mathbf{v}\,\Delta t, \qquad \text{if } p_y < y_{floor}:\ v_y \leftarrow -0.32\,v_y.$$

When the bounce speed drops below 0.35 m/s, the product lies on its side and waits. Pushing the flap rotates it inward around its top hinge, and the item goes into the player's bag. Any change falls as coins into the coin cup. An empty slot is refilled automatically after 40 s.

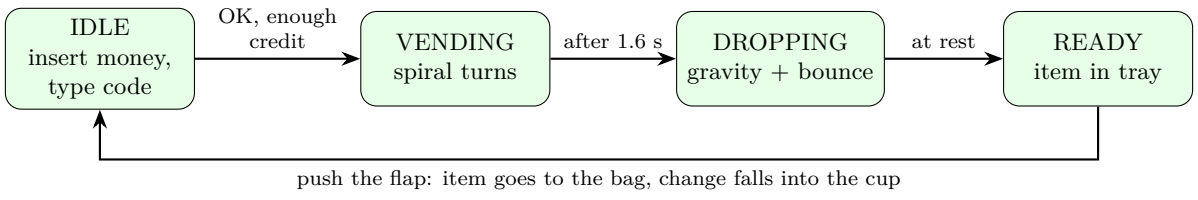


Figure 13: Vending machine state machine.



(a) Code B1 typed, credit enough **(b)** Spiral pushes the can out **(c)** "Take your item"

Figure 14: Buying an energy drink. The LCD, the price tags and the key labels are real text drawn on the machine.

7.2 Products

Table 10: Products and their shapes

Product	Built from
Can	cylinder body (.064, .11, .064) + colour band cylinder + 2 thin metal rim cylinders
Bottle	cylinder body + truncated-cone shoulder + thin neck cylinder + cap cylinder + label cylinders
Chips bag	box body + flattened sphere "puff" + 2 crimp boxes + round logo (cylinder rotated $R_x 90°$)
Chocolate bar	box wrapper + narrower box band
Cookie box	box + stripe box + round cookie picture (cylinder)
Juice box	box carton + front label box + tilted straw cylinder



Figure 15: Products (can, bottle, juice box, chips bag, chocolate bar, cookie box): the whole object and its parts.

Table 11: 6 products: the 23 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cylinder	12		–	$0.064 \times 0.11 \times 0.064$
cube	9		–	$0.13 \times 0.15 \times 0.04$
truncated cone	1		–	$0.062 \times 0.04 \times 0.062$
sphere	1		–	$0.125 \times 0.13 \times 0.06$

7.3 ATM

Table 12: Main parts of the ATM

Part	Shape	Size / transformation
Plinth, lower cabinet, upper fascia	cubes	$(1.0, .08, .72)$, $(.96, .92, .66)$, $(.96, .80, .50)$
Header with light strip	cubes	$(1.04, .24, .72)$ at $y = 1.96$; emissive strip under it
Privacy wings	cubes	$(.04, .62, .30)$ at $x = \pm 0.5$
Keypad shelf	cube in a child node	node $T(0, 1.035, 0.40) \cdot R_x(12°)$ so the keypad tilts towards the user
16 keypad keys	cubes	$(.062, .018, .042)$ on the tilted node; CANCEL red, CLEAR yellow, ENTER green
Screen bezel + screen	cube + text canvas	0.40×0.30 m screen drawn with rectangles and text
8 side keys	cubes	$(.04, .03, .016)$ at $x = \pm .245$, four rows
Card, cash, receipt	cubes	card slides along z, 4 bills slide out, receipt rotates $R_x(25°a)$
Camera dome	hemisphere	rotated $R_x(90°)$ above the screen



Figure 16: ATM: the whole object and its parts. The keypad shelf is one child node, tilted by $R_x(12°)$.

Table 13: ATM: the 49 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	38	+3 (glows)	–	$0.54 \times 0.4 \times 0.02$
cube	10		Brushed metal	$0.96 \times 0.92 \times 0.66$
hemisphere	1		–	$0.05 \times 0.025 \times 0.05$

7.4 Coffee machine, TV and aquarium

Coffee machine. It has a base box, a steel column, two black side panels, a head box, two nozzle cylinders, a drip tray (box + 6 thin bars), a bean hopper (a glass truncated cone turned upside down with $R_x(180°)$, so it is wider at the top), a beans cylinder, a lid, a glass water tank, three

round buttons (cylinders rotated $R_x(90°)$ with torus rings) and an LCD. The cup is an upside-down truncated cone with a torus handle. The coffee inside is a cylinder whose height is $0.07 \cdot fill$. The pouring stream is a thin cylinder that reaches from the nozzle to the coffee surface. Steam is made of small transparent spheres that rise, grow and fade.

TV. It has a wall bracket, a black body box, an emissive screen and a power LED. The four channels (news, nature, sports, weather) are 2D drawings made from rectangles and text on a canvas laid on the screen. Circles (heads, sun, trees, ball) are drawn as horizontal strips of width $2\sqrt{r^2 - y^2}$. When the TV is on, it also adds a flickering coloured point light to the room.

Aquarium. It has a wooden stand (box + 3 doors + handles), a sand box, rock spheres, a driftwood cylinder, water plants (cross cards that sway), a hood and a light strip. The water and glass are transparent boxes. Each fish is a scaled sphere body $(0.45s, 0.75s, 1.25s)$, a cone tail that wiggles with $R_y(25° \sin 9t)$, a wedge fin and two sphere eyes. Fish swim on ellipses $x = r_x \cos\varphi$, $z = r_z \sin\varphi$, and face their direction of motion, $yaw = \text{atan2}(\Delta x, \Delta z)$. "Feed the fish" drops food and the fish rise to it.



Figure 17: Coffee machine (brewing): the whole object and its parts. The steam puffs are small see-through spheres.

Table 14: Coffee machine: the 32 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	15	(see-through, glows)	–	$0.4 \times 0.04 \times 0.4$
cylinder	8		–	$0.25 \times 0.02 \times 0.25$
torus	4		–	$0.06 \times 0.06 \times 0.06$
truncated cone	2	(see-through)	–	$0.24 \times 0.18 \times 0.24$
cube	1		Brushed metal	$0.4 \times 0.56 \times 0.2$
cylinder	1		Rock	$0.17 \times 0.09 \times 0.17$
sphere	1	(glows)	–	$0.012 \times 0.012 \times 0.012$

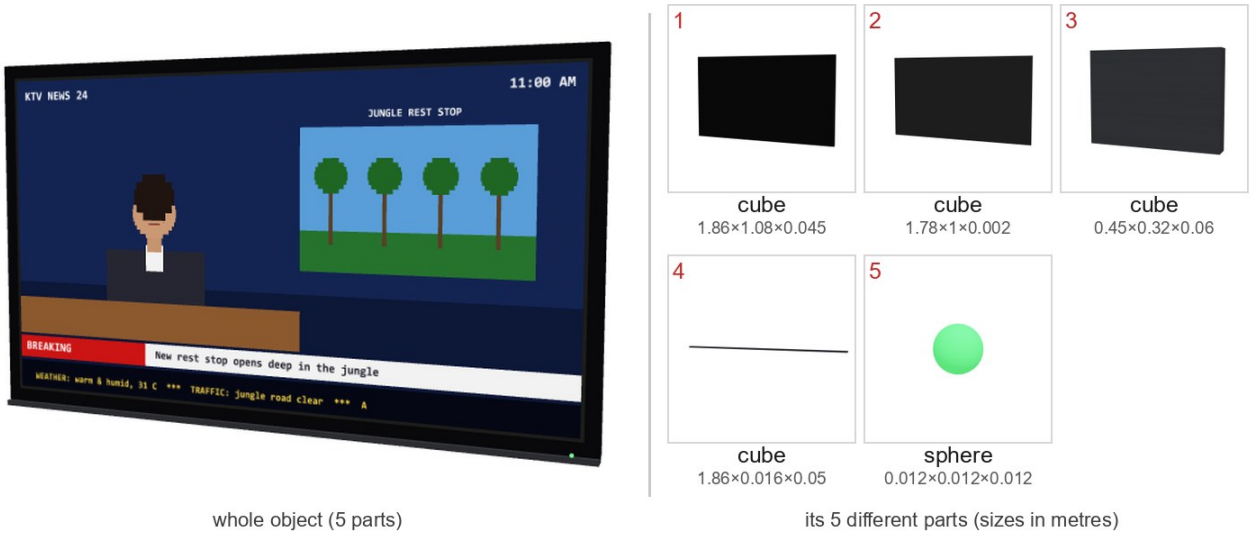


Figure 18: Television: the body is only four boxes; the picture is drawn as 2D shapes and text on the screen.

Table 15: Television (frame): the 5 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	2		Brushed metal	$0.45 \times 0.32 \times 0.06$
cube	2	(glows)	–	$1.86 \times 1.08 \times 0.045$
sphere	1	(glows)	–	$0.012 \times 0.012 \times 0.012$

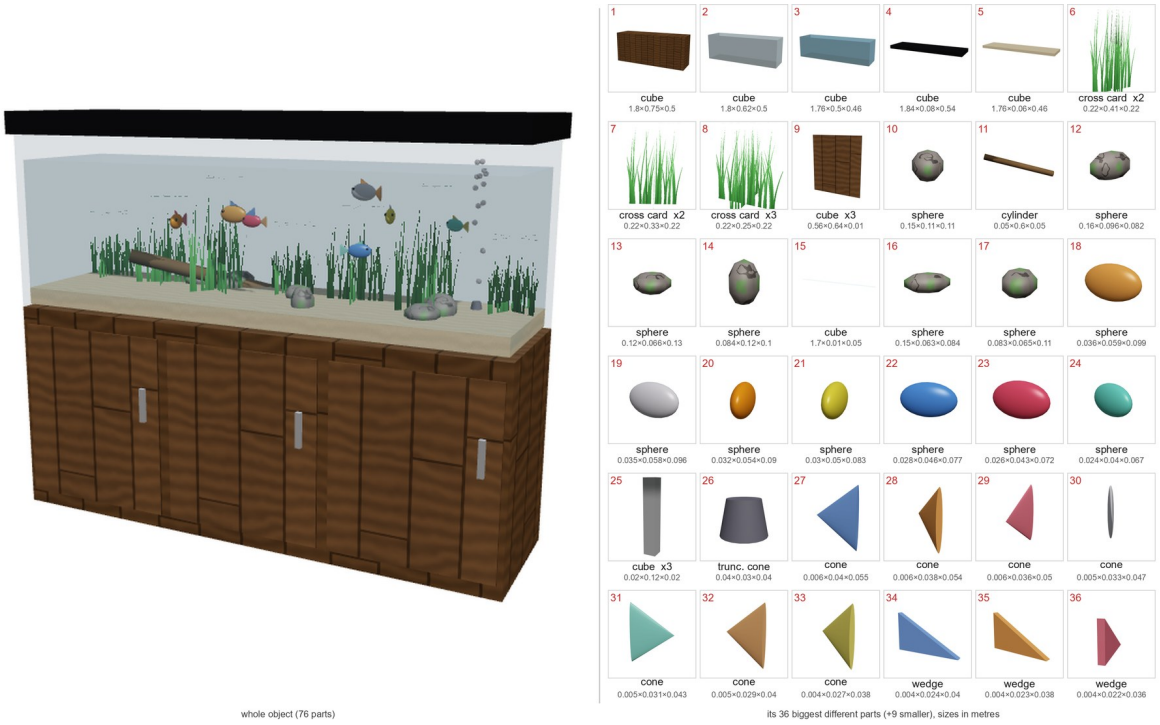


Figure 19: Aquarium: the whole object and its parts (stand, sand, rocks, plants, glass, water, fish, bubbles).

Table 16: Aquarium: the 76 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
sphere	35	(see-through)	–	$0.0357 \times 0.0594 \times 0.0991$
cube	7	(see-through, glows)	–	$1.8 \times 0.62 \times 0.5$
cross card	7		Grass blades	$0.22 \times 0.25 \times 0.22$
cone	7		–	$0.00634 \times 0.0396 \times 0.0555$
wedge	7		–	$0.00396 \times 0.0238 \times 0.0396$
sphere (low)	6		Rock	$0.0836 \times 0.119 \times 0.104$
cube	4		Wood planks	$1.8 \times 0.75 \times 0.5$
cube	1		Sand	$1.76 \times 0.06 \times 0.46$
cylinder (low)	1		Tree bark	$0.05 \times 0.6 \times 0.05$
truncated cone	1		–	$0.04 \times 0.03 \times 0.04$

7.5 Furniture and lamps

Table 17: Furniture made from basic shapes

Object	Construction
Sofa (n seats)	width $W = 0.72n + 0.32$; base box, back box; per seat: cushion box + front roll cylinder ($R_z 90°$) + back cushion box tilted $R_x(-10°)$; two arms (box + rolled cylinder $R_x 90°$); 4 wooden leg cylinders; throw pillows (boxes rotated in 3 axes)
Armchair	the same function with $n = 1$
Coffee table	top and shelf boxes, 4 metal leg boxes, magazines (thin boxes), fruit bowl (hemisphere with negative y scale) + spheres
Bookshelf	2 side boards, top, back, plinth, 5 shelves; each row of books is a plain box (page colour) with a quad in front carrying the procedural *books* texture. Each shelf uses two rows of different widths, and on every second shelf one quad is mirrored (negative x scale), so the pattern does not repeat
Wall clock	torus rim ($R_x 90°$), face cylinder, 12 ticks (each in a node $R_z(-30° i)$), hands $R_z(-30° h)$ and $R_z(-6° m)$. The hands show the *in-game* time
Pendant lamp	ceiling disc, cord cylinder, shade (truncated cone), diffuser disc, bulb sphere (emissive when on)
Street lamp	concrete base cylinder, tapered pole (truncated cone), arm cylinder $R_x 90°$, head (tapered box $R_x 180°$), glowing lens
Others	table lamps, floor lamp, potted plants (3 kinds), counters with cabinets, water cooler, benches, bins, stanchions with ropes, security cameras (they pan), smoke detectors, roof AC units, sign boards



Figure 20: Sofa (3 seats): the whole object and its parts.

Table 18: Sofa (3 seats): the 21 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	12		Fabric	$2.48 \times 0.22 \times 0.88$
cylinder	5		Fabric	$0.19 \times 0.88 \times 0.19$
cylinder	4		Wood planks	$0.05 \times 0.09 \times 0.05$

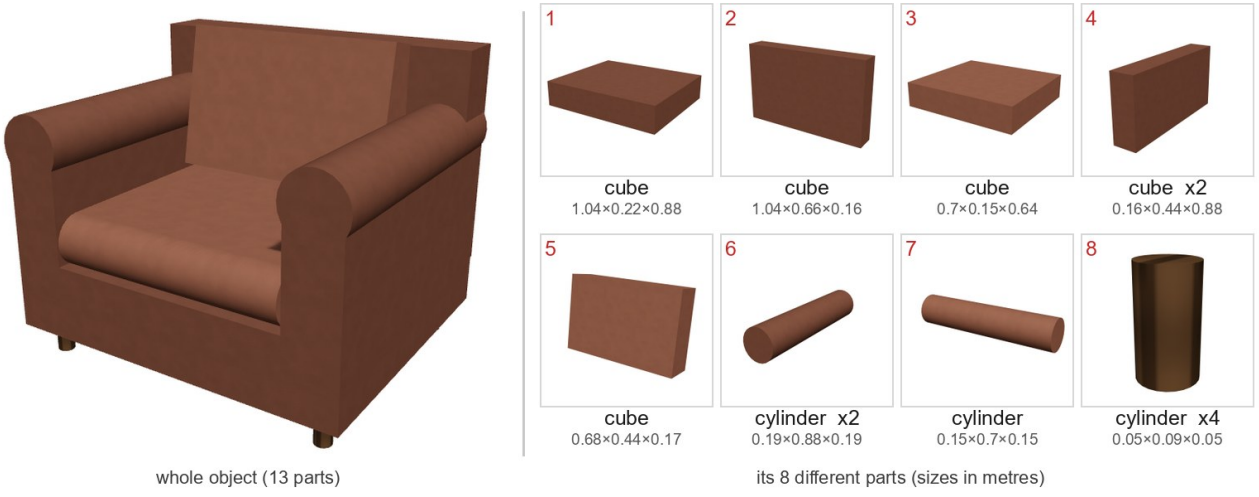


Figure 21: Armchair: the same sofa function with one seat.

Table 19: Armchair: the 13 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	6		Fabric	$1.04 \times 0.22 \times 0.88$
cylinder	4		Wood planks	$0.05 \times 0.09 \times 0.05$
cylinder	3		Fabric	$0.19 \times 0.88 \times 0.19$

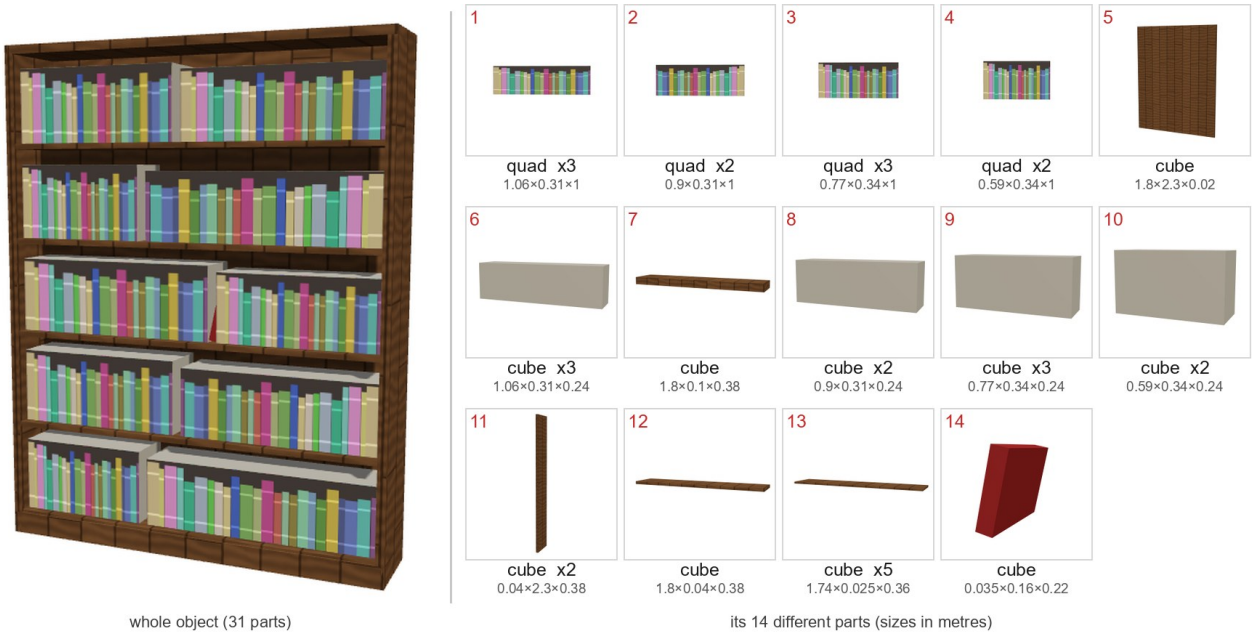


Figure 22: Bookshelf: boards and shelves, plus rows of books that are plain boxes with a *books* texture in front.

Table 20: Bookshelf: the 31 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	11		–	$0.595 \times 0.34 \times 0.24$
cube	10		Wood planks	$0.04 \times 2.3 \times 0.38$
quad	10		Books	$0.595 \times 0.34 \times 1$

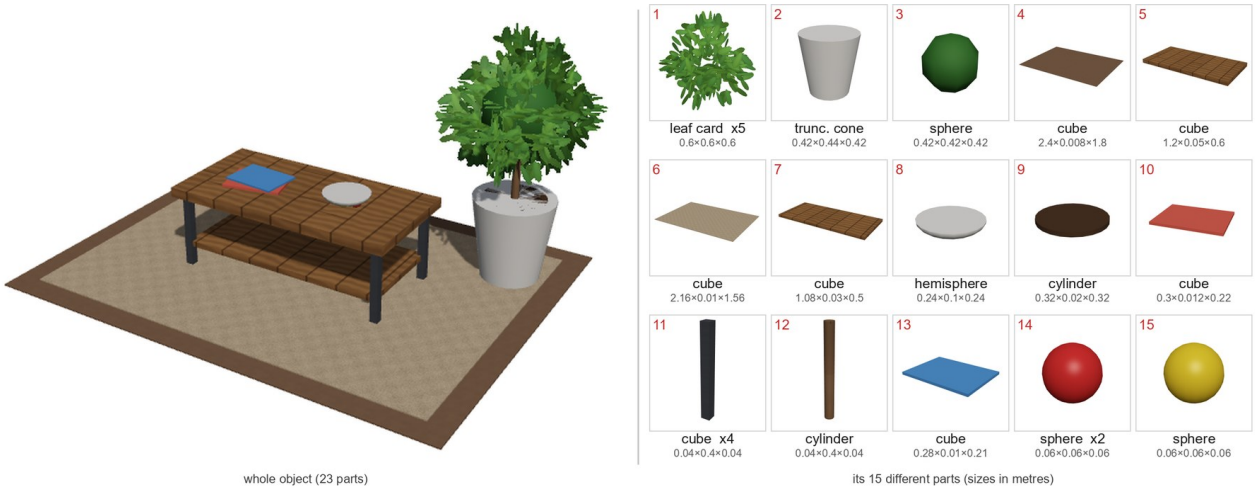


Figure 23: Table set: rug, coffee table with magazines and a fruit bowl, and a potted plant.

Table 21: Table set: the 23 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
leaf card	5		Leaf cluster	$0.6 \times 0.6 \times 0.6$
cube	4		Brushed metal	$0.04 \times 0.4 \times 0.04$
sphere	3		–	$0.06 \times 0.06 \times 0.06$
cube	2		Wood planks	$1.2 \times 0.05 \times 0.6$
cube	2		–	$0.3 \times 0.012 \times 0.22$
cube	1		Fabric	$2.4 \times 0.008 \times 1.8$
cube	1		Carpet	$2.16 \times 0.01 \times 1.56$
hemisphere	1		–	$0.24 \times 0.1 \times 0.24$
truncated cone	1		–	$0.42 \times 0.44 \times 0.42$
cylinder	1		–	$0.32 \times 0.02 \times 0.32$
cylinder	1		Tree bark	$0.04 \times 0.4 \times 0.04$
sphere (low)	1		–	$0.42 \times 0.42 \times 0.42$

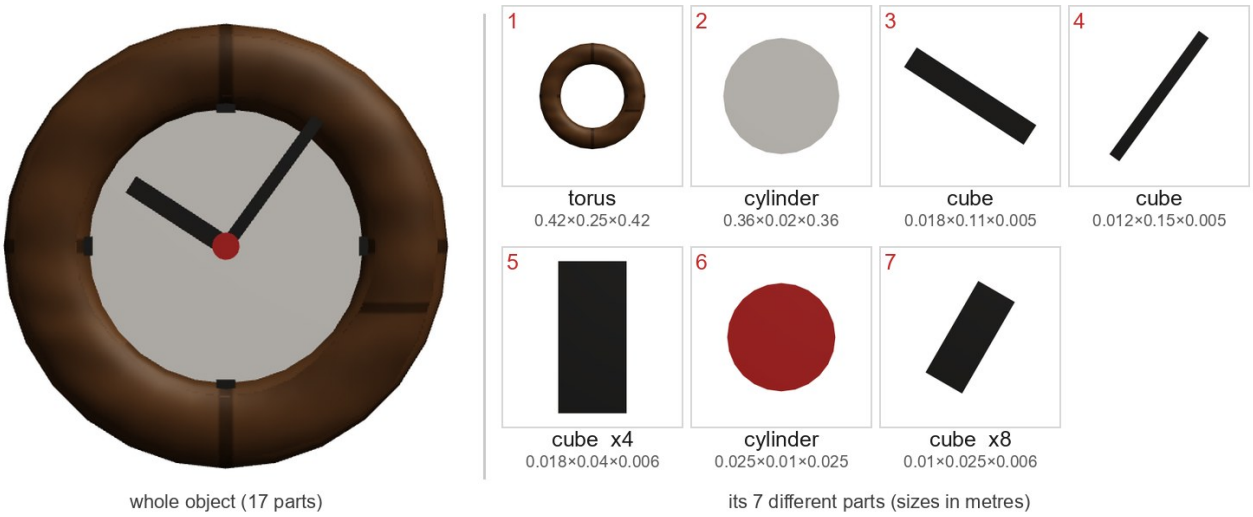


Figure 24: Wall clock: torus rim, face, 12 tick marks and the hands.

Table 22: Wall clock: the 17 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	14		–	$0.018 \times 0.04 \times 0.006$
cylinder	2		–	$0.36 \times 0.02 \times 0.36$
torus	1		Wood planks	$0.42 \times 0.25 \times 0.42$



Figure 25: Pendant lamp: ceiling disc, cord, shade, diffuser and the glowing bulb.

Table 23: Pendant lamp: the 5 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cylinder	2	(glows)	–	$0.42 \times 0.01 \times 0.42$
cylinder	1		Brushed metal	$0.14 \times 0.03 \times 0.14$
truncated cone	1		Brushed metal	$0.46 \times 0.26 \times 0.46$
sphere	1	(glows)	–	$0.11 \times 0.11 \times 0.11$

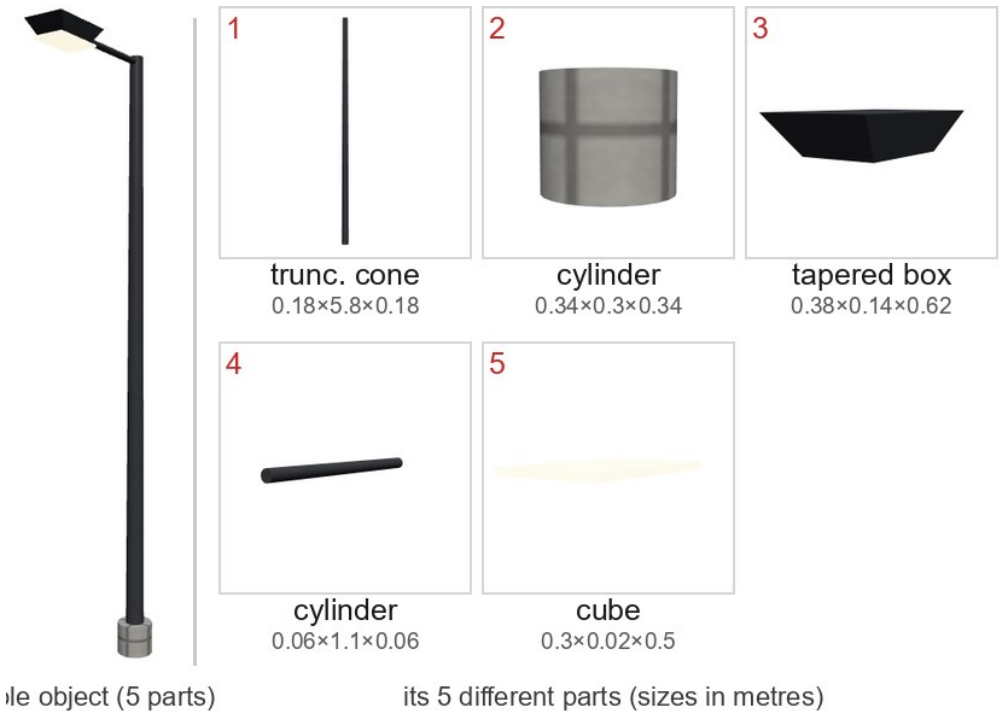


Figure 26: Street lamp: base, tapered pole, arm and the lamp head with its glowing lens.

Table 24: Street lamp: the 5 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cylinder	1		Concrete	$0.34 \times 0.3 \times 0.34$
truncated cone	1		Brushed metal	$0.18 \times 5.8 \times 0.18$
cylinder	1		Brushed metal	$0.06 \times 1.1 \times 0.06$
tapered box	1		Brushed metal	$0.38 \times 0.14 \times 0.62$
cube	1	(glows)	–	$0.3 \times 0.02 \times 0.5$

7.6 People

A person has about 50 parts (Figure 11). To keep things simple there are only two types, each with one fixed colour scheme. Both use the same hierarchy and the same code; only a few numbers and colours differ (Table 25). The player's own body is the man.

Table 25: The two types of people

	Man (type 0)	Woman (type 1)
Hair	short: hemisphere + sphere	long: hemisphere + a tall flat sphere behind the head
Top	blue shirt	red kurta (the hips are red too, no belt)
Lower body	dark trousers	light trousers + kurta hem (upside-down truncated cone)
Size	height $S = 1.0$, width $w = 1.0$	height $S = 0.95$, width $w = 0.94$

Table 26: Body parts of a person (local sizes in metres, w = width factor)

Part	Shape	Size	Placement
Pelvis + belt	cylinder, cube	$(.34w, .18, .21)$	pelvis node at $y = 0.93 - drop + bob$ (the woman has no belt)
Hip joint, thigh, knee, shin	sphere, truncated cone (upside down), sphere, truncated cone	thigh $(.17w, .45, .17)$, shin $(.135w, .42, .135)$	hip joint $(\pm.092w, -.03, 0)$, $R_x(\theta_{thigh})$; knee joint $T(0, -.45, 0)R_x(\theta_{knee})$. The cones are flipped with $R_x(180°)$, so legs are thicker at the top
Ankle, shoe, toe, sole	sphere, cube, cylinder, cube	$(.11, .07, .26)$	ankle node, toes point $+z$; rounded toe = cylinder $R_z(90°)$
Waist	cylinder	$(.30w, .22, .19)$	torso node $T(0, .08, 0)R_x(lean)$
Chest	tapered box $R_z(180°)$	$(.42w, .30, .23)$	upside down, so it is wide at the shoulders and narrow at the waist (V shape)
Shoulder line, joints, collar	cylinder $R_z 90°$, spheres, torus	joints .115	joints at $(\pm.215w, .425, 0)$, exactly where the arms start
Neck, head, jaw	cylinder, spheres	head $(.20, .245, .215)$	neck node, then $R_y(\text{headYaw})R_x(\text{headPitch})$
Face	spheres + boxes		eyes (white + pupil), nose, ears, eyebrows, mouth; glasses = 2 tori
Hair	hemisphere (+ sphere)		2 styles (short, long)
Upper arm, elbow, forearm, hand, thumb	truncated cones (flipped), spheres	$(.11, .29, .11)$, $(.088, .26, .088)$	shoulder $R_x(\theta_s)R_z(\pm out)$, elbow $R_x(\theta_e)$; arms taper towards the wrist; a held item (can, cup, snack, phone) is a child of the right hand

Animation. Only the joint angles change. For walking with phase φ:

$$\theta^{L,R}_{thigh} = \mp 26° \sin\varphi, \quad \theta^{L}_{knee} = 38° \max(0, \cos\varphi) + 4°, \quad \theta^{L,R}_{shoulder} = \pm 22° \sin\varphi, \quad bob = 0.035\,(1 - |\sin\varphi|).$$

Sitting lowers the pelvis by 0.44 m, turns the thighs $-88°$ and bends the knees $88°$. Driving, pressing buttons and bending down are other fixed poses. Changes between poses are smooth, because every

angle is blended linearly over 0.35 s.



Figure 27: The man (type 0, walking pose): the whole object and its parts. Every joint is a sphere, every limb a truncated cone.

Table 27: Man (walking): the 49 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
sphere	16		–	$0.207 \times 0.21 \times 0.17$
cube	8		–	$0.335 \times 0.035 \times 0.205$
sphere	8		Fabric	$0.16 \times 0.16 \times 0.17$
truncated cone	6		Fabric	$0.17 \times 0.45 \times 0.17$
cylinder	3		Fabric	$0.34 \times 0.18 \times 0.21$
cylinder	3		–	$0.095 \times 0.09 \times 0.095$
truncated cone	2		–	$0.088 \times 0.26 \times 0.088$
tapered box	1		Fabric	$0.42 \times 0.3 \times 0.23$
torus	1		Fabric	$0.2 \times 0.12 \times 0.17$
hemisphere	1		–	$0.215 \times 0.25 \times 0.232$



Figure 28: The woman (type 1): the same parts, plus long hair and the kurta hem.

Table 28: Woman (standing): the 50 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
sphere	16		–	$0.204 \times 0.342 \times 0.123$
sphere	8		Fabric	$0.143 \times 0.152 \times 0.162$
truncated cone	7		Fabric	$0.357 \times 0.342 \times 0.285$
cube	7		–	$0.104 \times 0.0665 \times 0.247$
cylinder	3		Fabric	$0.304 \times 0.171 \times 0.2$
cylinder	3		–	$0.0902 \times 0.0855 \times 0.0902$
truncated cone	3		–	$0.0836 \times 0.247 \times 0.0836$
tapered box	1		Fabric	$0.375 \times 0.285 \times 0.218$
torus	1		Fabric	$0.19 \times 0.114 \times 0.162$
hemisphere	1		–	$0.211 \times 0.247 \times 0.224$



(a) Woman, standing with a cup **(b)** Man, walking pose **(c)** Sitting on a chair

Figure 29: The two types and their poses, all from the same hierarchical model.

7.7 Cars

There is one car model, a sedan of $4.4 \times 1.78 \times 1.42$ m, painted in one of three colours: red, blue or white. The colour is the only difference, so all three use the same vertices and matrices (the tyre and

door examples in Section 6.3 are from this car).

Table 29: Parts of a car

Part	Construction
Body	a front block and a rear block (boxes), a thin passenger-side skin and a floor in between, rounded nose (cylinder lying along x, $R_z 90°$), bumpers, underbody. The driver-door area is left *open*, so an opened door shows the inside of the car, not a second door
Bonnet	wedge that slopes down towards the front; boot lid box
Cabin	roof box + tinted *transparent* glass (windscreen, rear window, 4 side windows), so the driver is visible. Each window is a thin parallelogram (`panel`) plus, where needed, a triangle (`tri`, a wedge); both matrices are built straight from the corner points (Section 6.3). The windscreen and rear window get painted corner triangles, the front side windows a small quarter glass with a divider bar, and the rear side windows a painted C-pillar triangle
Pillars	thin boxes stretched between two points by a "beam" helper: the box's y axis is turned onto $\mathbf{b}-\mathbf{a}$ with the rotation $\mathbf{y}\times\mathbf{d}$ by atan2($\lvert\mathbf{y}\times\mathbf{d}\rvert$, $\mathbf{y}\cdot\mathbf{d}$)
Interior	seats, rear bench, dashboard, steering wheel (torus $R_x 70°$)
Lights	head lights (emissive at night + real spot lights), tail and brake lights, reverse lights, blinking orange indicators
Wheels (4)	dark wheel well, tyre cylinder $(2R, .22, 2R)$ $R_z 90°$, chrome rim, 3 spokes, hub; node $R_y(steer)$ for the front wheels, then $R_x(spin)$
Driver door	child node $T(hinge)\cdot R_y(\theta_{door})$ holding the outer panel, inner trim, arm rest, handle and *its own window*. The window corners are the same points as in the car minus the hinge position, so the glass swings out with the door. It opens to 70° when the NPC gets in or out

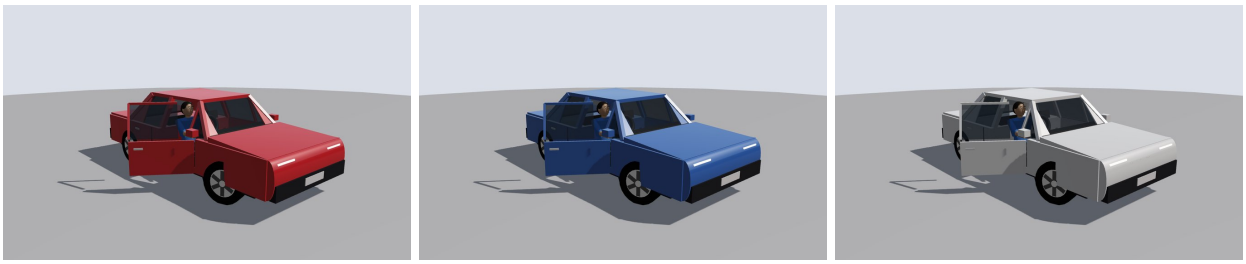
The wheels turn with θ_{spin} += $\Delta s/R$, where Δs is the distance driven. The steering angle follows the car's turning rate, $\delta = \arctan(L_{wb}\,\Delta\psi/\Delta s)$, where L_{wb} is the wheel base and $\Delta\psi$ is the change of heading.



Figure 30: The car: the whole object (with its driver) and the car's parts (door open 55°, front wheels steered 18°).

Table 30: Car + driver (any colour): the 144 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cube	76	+4 (see-through, glows)	–	$1.78 \times 0.556 \times 1.42$
cylinder	20		–	$0.545 \times 1.74 \times 0.24$
sphere	16		–	$0.207 \times 0.21 \times 0.17$
wedge	9	(see-through)	–	$1.73 \times 0.08 \times 1.15$
sphere	8		Fabric	$0.16 \times 0.16 \times 0.17$
truncated cone	6		Fabric	$0.17 \times 0.45 \times 0.17$
cylinder	3		Fabric	$0.34 \times 0.18 \times 0.21$
truncated cone	2		–	$0.088 \times 0.26 \times 0.088$
torus	1		–	$0.38 \times 0.38 \times 0.38$
tapered box	1		Fabric	$0.42 \times 0.3 \times 0.23$
torus	1		Fabric	$0.2 \times 0.12 \times 0.17$
hemisphere	1		–	$0.215 \times 0.25 \times 0.232$



(a) Red **(b)** Blue **(c)** White

Figure 31: The one car model in its three colours (driver door opened to 55°).

7.8 The building

Walls are made by one function, `buildWall()`. It takes a straight wall and a list of openings (doors and windows), and splits the wall into pieces (Figure 32):

1. full-height pieces between the openings,
2. for each opening, a piece *below* it (a window sill wall) and a piece *above* it (lintel),
3. for windows: a transparent glass pane, an aluminium frame (4 bars + a middle bar) and a marble sill,
4. for doors: frame trims.

Each piece has two half-thickness layers. The outside layer uses *brick*, and the inside layer uses the room's *plaster* colour (beige lounge, light blue ATM room, sage green library). Wooden skirting boards run along the inside floor line. Floors are tiles (lounge), carpet (ATM room) and wood (library). The ceilings and roof (with parapet, AC units and the sign) are drawn per frame, so the roof can be hidden in the cut-away view. The automatic sliding glass doors slide sideways by $x = \pm(0.6 + 1.15 \cdot open)$ while someone stands within 2.2×2.8 m of them. While they are closed they also block the player.

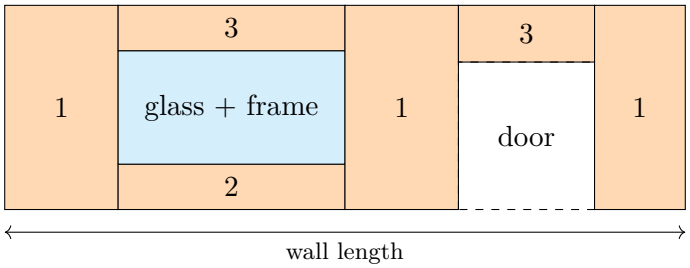


Figure 32: A wall with one window and one door: (1) full pieces, (2) below the window, (3) above the opening.

7.9 The jungle

Tree species.

- *Rainforest tree*: low-poly cylinder trunk (6.5–9 m), 4 **buttress roots** (wedges rotated $R_y(90°i)$), 3 branches (cylinders in nodes $R_y(yaw)R_x(45–60°)$), and a crown made mostly of **leaf cards**: 15–19 alpha-tested cards, each turned randomly in all three axes and placed on an ellipsoid shell, plus 3 small dark core blobs inside that give volume. Every card gets a slightly different green. Some branches carry hanging vines.
- *Coconut palm*: a chain of 7 trunk segments. Each segment is a child of the previous one with an extra $R_x(3–7°)$, so the trunk bends smoothly. 9 fronds, each made of two quads (the second is a child of the first and droops by another 30–45°), and 4 coconuts.
- *Emergent giant*: a 14–18 m trunk, 5 large buttresses, 4 branches and a wide layered crown.
- *Bamboo clump*: 9–15 tilted culms (thin cylinders) with node rings and leaf cards.
- *Banana plant*: 1–3 pseudo-stems with 5–8 long leaves (spheres flattened to $(0.55, 0.04, 1.9)$), sometimes with a banana bunch.
- *Undergrowth*: bushes (one small dark core + 5–7 leaf cards, some with flowers), ferns (5 arched frond quads + a cross card), grass tufts and flower cards, rocks (1–3 rotated, squashed low spheres with a rock texture) and fallen logs (horizontal cylinder + moss blobs + mushrooms made of a cylinder and a hemisphere).

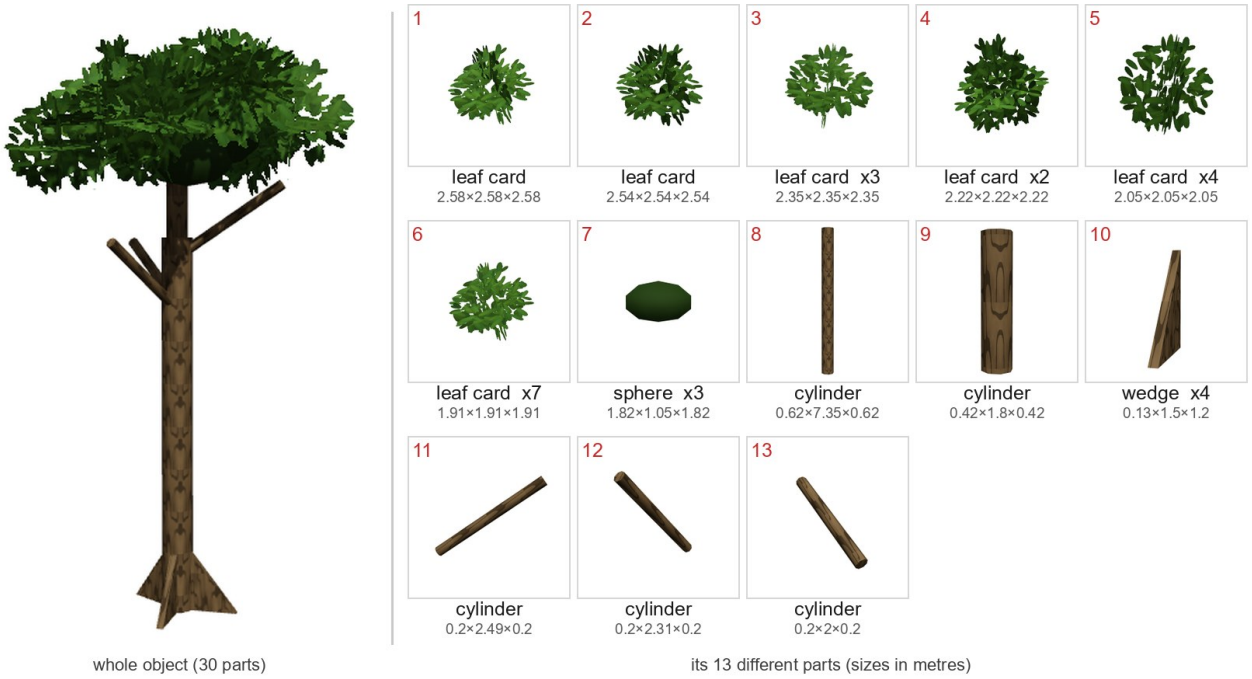


Figure 33: Rainforest tree: trunk, buttress roots (wedges), branches, dark core blobs and leaf cards.

Table 31: Rainforest tree: the 30 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
leaf card	18		Leaf cluster	$2.58 \times 2.58 \times 2.58$
cylinder (low)	5		Tree bark	$0.62 \times 7.35 \times 0.62$
wedge	4		Tree bark	$0.13 \times 1.5 \times 1.2$
sphere (low)	3		–	$1.82 \times 1.05 \times 1.82$

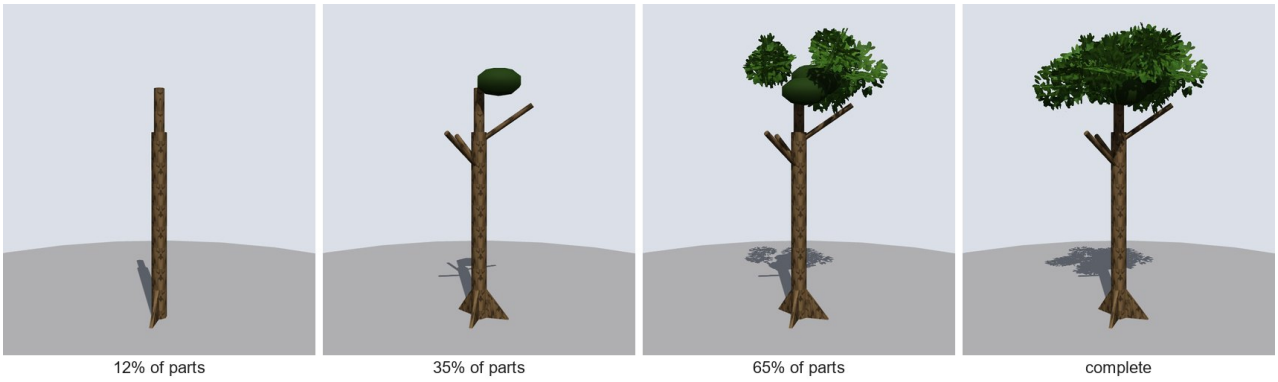


Figure 34: Rainforest tree build stages: trunk, buttress roots, branches, then the core blobs and the leaf cards.



Figure 35: Coconut palm: 7 chained trunk segments with rings, fronds made of two quads each, and coconuts.

Table 32: Coconut palm: the 37 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
quad	18		Palm frond	$1 \times 1.91 \times 1$
cylinder (low)	7		Tree bark	$0.36 \times 1.44 \times 0.36$
cylinder (low)	7		–	$0.403 \times 0.08 \times 0.403$
sphere (low)	5		–	$0.55 \times 0.6 \times 0.55$

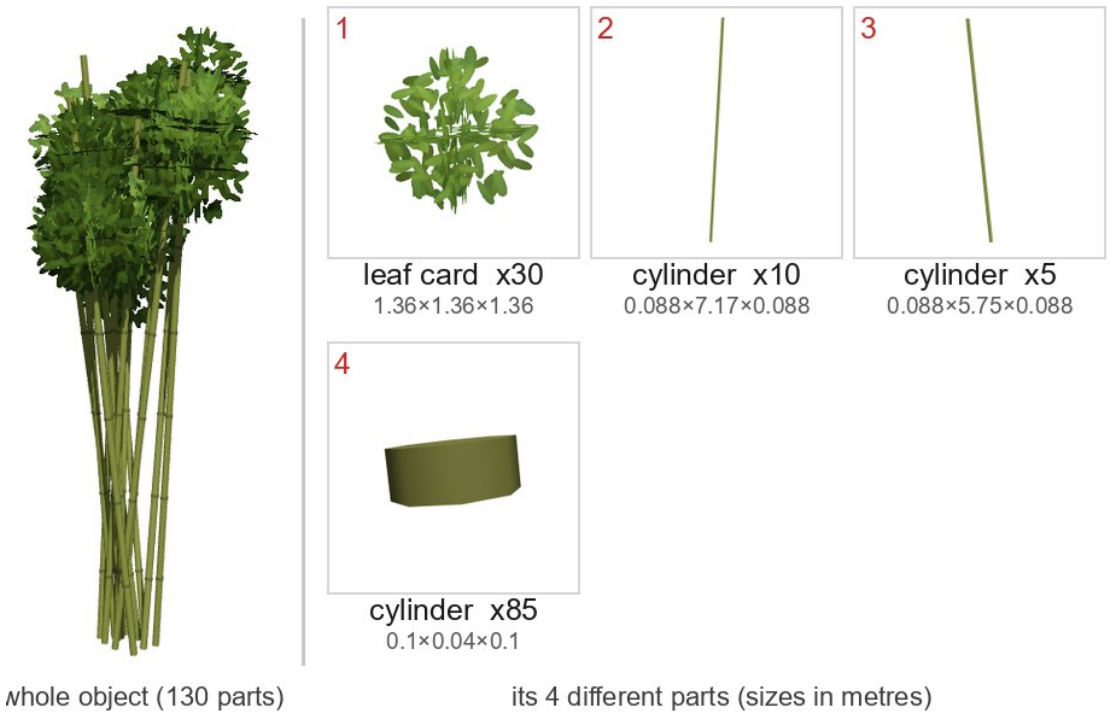


Figure 36: Bamboo clump: tilted thin cylinders (culms) with node rings and leaf cards.

Table 33: Bamboo clump: the 130 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
cylinder (low)	100		–	$0.088 \times 5.75 \times 0.088$
leaf card	30		Leaf cluster	$1.36 \times 1.36 \times 1.36$

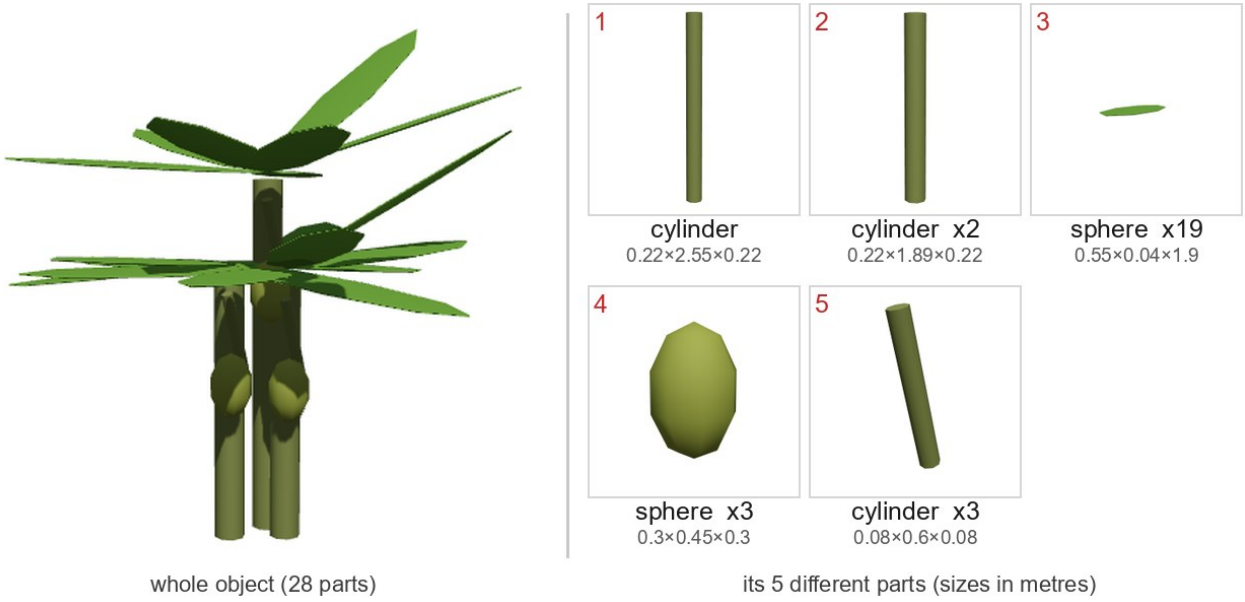


Figure 37: Banana plant: stems (cylinders) and long leaves (flattened spheres).

Table 34: Banana plant: the 28 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
sphere (low)	22		–	$0.55 \times 0.04 \times 1.9$
cylinder (low)	6		–	$0.22 \times 1.89 \times 0.22$

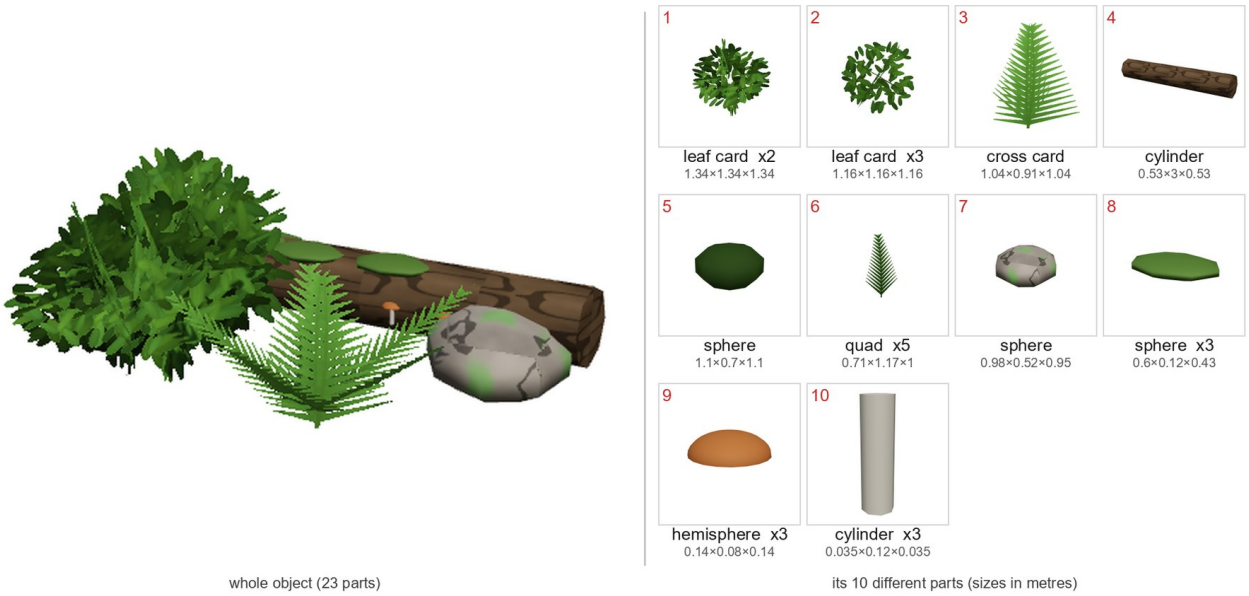


Figure 38: Undergrowth: a bush, a fern, a rock and a fallen log with moss and mushrooms.

Table 35: Bush+fern+rock+log: the 23 parts grouped by shape and texture (size = the largest part of the group, in metres)

Shape	No.	Colours	Texture	Largest size (s_x, s_y, s_z)
leaf card	5		Leaf cluster	$1.34 \times 1.34 \times 1.34$
quad	5		Fern	$0.715 \times 1.17 \times 1$
sphere (low)	4		–	$1.1 \times 0.7 \times 1.1$
cylinder (low)	3		–	$0.035 \times 0.12 \times 0.035$
hemisphere	3		–	$0.14 \times 0.08 \times 0.14$
cross card	1		Fern	$1.04 \times 0.91 \times 1.04$
sphere (low)	1		Rock	$0.983 \times 0.522 \times 0.946$
cylinder (low)	1		Tree bark	$0.532 \times 3 \times 0.532$

Placement. The world (280×280 m) is covered with a *jittered grid* of 6.6 m cells. Each cell gets one random point. Points inside the cleared area (building, lot and road with a margin) are skipped. A random number then picks the species: 40% rainforest tree, 18% palm, 7% giant, 8% bamboo, 11% banana, 8% bush, and 8% nothing. Every cell also gets 1–5 ferns or bushes and 4–12 grass or flower tufts (more at the edge of the clearing). There is a 12% chance of a rock and a 5% chance of a log. In total the jungle has 795 broad-leaf trees, 279 palms, 132 bamboo clumps, 1,503 bushes, 2,193 ferns, 4,783 grass tufts, 199 rocks and 86 logs. Every trunk also gets a circle collider.

Terrain. The ground is one displaced grid. It is flat in the clearing and along the road, and becomes gently hilly further away:

$$H(x,z) = 14\,\big(\mathrm{fbm}(0.018x,\,0.018z) - 0.45\big) + 1.2\,\mathrm{fbm}(0.07x,\,0.07z)$$

$$h(x,z) = \max\Big(-1.5,\;\; H(x,z)\cdot \mathrm{smoothstep}(0,30,d)\cdot \mathrm{smoothstep}\big(5,14,|z - z_{road}(x)|\big)\Big)$$

Here d is the distance outside the clearing. The road is straight through the lot ($z = 16$) and curves in the jungle. With $t = |x| - 30$:

$$z_{road}(x) = 16 + 7\sin(0.035\,t)\,\mathrm{sign}(x)\cdot \mathrm{smoothstep}(0,25,t).$$

The smoothstep factor makes the bend start gently (its slope is 0 at $t = 0$). Without it the road had a sudden 14° kink at $|x| = 30$, and a wedge-shaped gap opened between two road boxes there. The road is built from 2 m long boxes turned to follow the curve; neighbouring boxes overlap by 40 cm, which covers the small gap on the outer side of every bend. Figure 1 shows the clearing in the jungle.

8 Lighting and Shading

8.1 Illumination model

The **illumination model** says how much light leaves a point of a surface. We use the **Phong illumination model** with Blinn's half vector (the *Blinn–Phong* model). It adds three kinds of reflected light:

Table 36: The three terms of the Phong illumination model

Term	Formula	Meaning
Ambient	$I_a\,k_a$	light that has bounced around the scene; it reaches every surface, even in shadow, so nothing is fully black
Diffuse	$I_l\,k_d\,\max(\mathbf{N}\cdot\mathbf{L}, 0)$	**Lambert's cosine law**: a surface facing the light is bright, one at a grazing angle is dark. It looks the same from every viewing direction (matte paint, cloth, wood)
Specular	$I_l\,k_s\,\max(\mathbf{N}\cdot\mathbf{H}, 0)^n$	the shiny highlight. It moves when the camera moves. A large shininess n (chrome: 128) gives a small sharp spot, a small n (fabric: 8) a wide dull one

Here $\mathbf{N}$ is the unit normal, $\mathbf{L}$ the unit vector to the light, $\mathbf{V}$ the unit vector to the camera and $\mathbf{H} = \text{normalize}(\mathbf{L}+\mathbf{V})$ the half vector. Phong's original model uses the mirror direction $\mathbf{R} = 2(\mathbf{N}\cdot\mathbf{L})\mathbf{N}-\mathbf{L}$ and $(\mathbf{R}\cdot\mathbf{V})^n$; Blinn's $(\mathbf{N}\cdot\mathbf{H})^n$ gives almost the same picture, is cheaper and has no sharp cut-off at wide angles. For one light the classic equation is

$$I = I_a k_a + I_l\big(k_d \max(\mathbf{N}\cdot\mathbf{L}, 0) + k_s \max(\mathbf{N}\cdot\mathbf{H}, 0)^n\big).$$

Our shader uses the same equation, summed over all lights, with a few extra factors. For one light with direction $\mathbf{L}$:

$$C = \underbrace{\text{mix}(A_{ground}, A_{sky}, \tfrac{N_y+1}{2})\, k_d}_{\text{ambient}} + \sum_{lights} \Big(k_d \max(\mathbf{N}\cdot\mathbf{L}, 0) + k_s(\mathbf{N}\cdot\mathbf{H})^n\Big)\, I\, a(d)\, c\, s + E.$$

Here k_d is the surface colour (times the texture), k_s is the specular strength, n is the shininess, I is the light colour times its intensity, $a(d)$ is the attenuation, c is the spot cone factor, s is the shadow factor and E is the emission (screens, LEDs, bulbs). The ambient part uses *hemisphere* lighting: surfaces facing up get the sky colour and surfaces facing down get the ground colour. Inside the building the ambient part is multiplied by 0.45, because the roof blocks the sky. Leaves and grass use soft "wrap" lighting, $0.6\,\mathbf{N}\cdot\mathbf{L} + 0.4$, and are lit from both sides. When you look towards the sun through leaves, a *translucency* term $0.45\, k_d\, I\, s\, \max(-\mathbf{V}\cdot\mathbf{L}, 0)^3$ makes them glow yellow-green, like real thin leaves. Leaf cards are also a little darker at their base (a cheap kind of ambient occlusion).

Attenuation uses a smooth window, so every light really ends at its range r and can be skipped beyond it:

$$a(d) = \frac{\big[\text{clamp}(1-(d/r)^4,\, 0,\, 1)\big]^2}{1+d^2}, \qquad c = \text{smoothstep}\big(\cos\theta_{out},\ \cos\theta_{in},\ -\mathbf{L}\cdot\mathbf{D}\big) \text{ for spot lights.}$$

8.2 Shiny materials: polished metal and glossy paint

Every material has its own k_s and shininess n: fabric, plaster and bark are almost matte ($k_s \approx 0.05$, $n \approx 8$), wood and leather are a little shiny, marble and the floor tiles are glossy ($n \approx 90$). Two kinds of material get an extra term, because real metal and lacquer mostly show *what is around them*:

- **Polished metal (chrome)**: the chrome sculpture next to the entrance, the waste bins, the canopy posts, the car wheel rims and handles. The diffuse part is made weak ($0.25\, k_d$), the highlight is strong and has the metal's colour, and the surface reflects its surroundings.
- **Glossy lacquer (car paint)**: normal coloured paint with a clear coat on top. The clear coat adds a sharp white highlight and a reflection that becomes stronger at grazing angles.

The reflected direction is $\mathbf{R} = \text{reflect}(-\mathbf{V}, \mathbf{N}) = 2(\mathbf{N}\cdot\mathbf{V})\mathbf{N} - \mathbf{V}$. No real reflection is traced; instead the colour seen in direction $\mathbf{R}$ is a simple painted surrounding $E(\mathbf{R})$: bright sky above, a thin dark band of jungle at the horizon, sunlit ground below, and the glow of the sun (inside the building: ceiling, walls and floor). The amount of reflection uses Schlick's Fresnel approximation, $F = (1-\mathbf{N}\cdot\mathbf{V})^5$, which is large when you look along the surface:

$$C_{metal} = \text{Phong}(0.25\, k_d,\ 3k_d) + E(\mathbf{R})\, k_d\, (0.85{+}0.15F), \qquad C_{paint} = \text{Phong}(k_d,\ 2.5) + E(\mathbf{R})\, (0.08{+}0.75F).$$

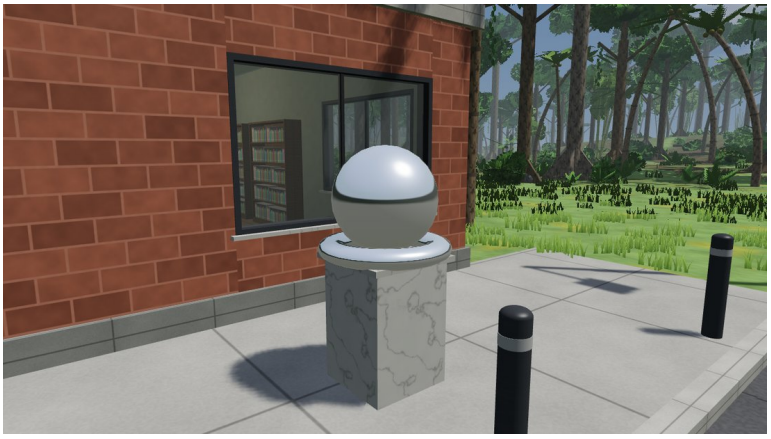


Figure 39: The chrome sculpture: the sharp specular highlight of the sun (top right), the sky and the dark tree line reflected around the middle, and the ground reflected below. The marble pedestal shows a softer, wider highlight.

8.3 Light types and where they are shown

Table 37: Light types in the scene, and the pictures in this report where each one can be seen

Type	How it works	Where in the scene	Shown in
Directional	same direction **L** everywhere, no distance fall-off (the source is very far away)	the **sun** by day and the **moon** by night (one light that follows the clock)	Figure 6: sunrise, day, sunset, night
Point	shines in all directions from one position, gets weaker with distance $a(d)$	lounge pendant bulbs (4), library pendants (3), ATM tube lights (2), vending machine interiors (3), ATM screen, aquarium hood, TV glow (flickers with the picture), 8 parking lamps and road lamps	Figure 40 (pendants), Figure 41 (street lamps), Figure 52 (TV glow)
Spot	a point light limited to a cone (inner and outer angle)	downlights over the vending machines, TV wall and coffee counter, the reading floor lamp, porch lights, **car headlights** (moving spot lights)	Figure 14 (vending downlight), Figure 41 (reading lamp, headlights)
Emissive	the surface glows by itself; it lights nothing else	glowing bulbs and tubes, LEDs, screens, signs, fireflies, the marquee	Figure 40, Figure 25
Ambient	a constant base light, here a mix of sky colour (from above) and ground colour (from below)	everywhere; it changes with the time of day	Figure 40(b): with the lamps off only ambient, moon and glow remain

Every light belongs to a *group*: lounge, spots, ATM room, library, outdoor or machines. Groups can be switched with the F1–F4 keys or with the real **light switches** on the walls, which you can look at and press. The lamp models check their group, so a bulb only glows when its light is on. The outdoor group switches on automatically at dusk.

8.4 Shading: where the equation is evaluated

The illumination model gives the light at *one* point. The **shading model** decides at which points it is computed and how the colour is filled in across a triangle:

Table 38: The three classic shading methods

Method	How	Look
Flat shading	one normal per triangle, the equation is used once, the whole triangle gets one colour	faceted, every triangle visible
Gouraud shading	the equation is used at the 3 vertices (vertex shader), the colours are blended across the triangle	smooth, but highlights in the middle of a triangle are lost or smeared
Phong shading (ours)	the *normal* is blended across the triangle, and the equation is used at every pixel (fragment shader)	smooth, sharp correct highlights

We use **Phong shading**. The vertex shader sends the world position and the normal $N\,\mathbf{n}$ (with the normal matrix $N = (M^{-1})^T$) to the fragment shader; the GPU interpolates them across the triangle, and the fragment shader normalises the normal again and evaluates the lighting. Whether a shape looks smooth or faceted depends on its *vertex normals*, not on the shading method: a cube has separate vertices for each face with the face normal, so each face is flat and the edges are sharp; a sphere or cylinder has one normal per vertex pointing straight out, so it looks round even with only 20 or 24 sides (see the shapes in Section 5.1). In Basic mode the same Phong shading is used, only without textures, shadows and fog.

8.5 Choosing lights every frame

The shader accepts up to 16 point and 8 spot lights. Every frame the light manager:

1. drops lights that are switched off, too far away (> 45 m beyond their range), or whose sphere of influence is outside the camera frustum,
2. keeps the closest ones,
3. sorts them by **region**: ATM room, lounge, library or outdoors.

Each pixel works out its own region from its position and loops *only* over the lights of that region. This has two benefits. Light no longer leaks through walls into the next room, and pixels do much less work. In the lounge view this change raised the frame rate from 53 to 63 FPS.



(a) Night, lamps on



(b) Night, lamps switched off (only machine glow and moonlight)

Figure 40: The same room with and without its bulbs and spot lights.



(a) Library at night: warm pendants and a reading lamp



(b) Sodium street lamps and car lights

Figure 41: Different kinds of light at night.

9 Day and Night

A 24-hour clock runs at 1 game hour per 20 s by default (keys change the speed, pause it, or jump between day and night). The sun's direction comes from the hour:

$$a = \frac{hour - 6}{12}\pi, \qquad \mathbf{s} = \text{normalize}(0.95\cos a,\ \sin a,\ 0.42), \qquad \mathbf{m} = \text{normalize}(-0.8\,s_x,\ -s_y,\ 0.3).$$

So the sun rises in the east ($+x$) at 6:00, is highest at noon and sets at 18:00, and the moon is on the opposite side. All other values are blended from the sun height s_y: the sky zenith and horizon colours (with an orange glow near sunrise and sunset), the sun colour (warm when low), the ambient light, the fog colour and density, the cloud colour and the exposure. The night factor $1 - \text{smoothstep}(-0.14,\ 0.10,\ s_y)$ shows the stars and moon, turns on the street lamps and car headlights, and wakes up the fireflies. Figure 6 shows the same view at four times of day.

10 Textures

10.1 Procedural textures

All 25 textures are created at start-up from *value noise*. Random values are placed on a grid and smoothly interpolated between grid points. Several octaves are then added together (fractal Brownian motion):

$$\text{fbm}(\mathbf{p}) = \frac{\sum_{i=0}^{k-1} 0.5^i\ \text{noise}(2^i\mathbf{p})}{\sum_{i=0}^{k-1} 0.5^i}.$$

The noise wraps around at the texture border, so the textures tile without seams. Patterns are built on top of it: rows of bricks with mortar, wooden planks with grain $\sin(6v + \text{noise})$, tiles with grout, marble veins, woven fabric, bark ridges, and also *cut-out* textures (leaves, grass blades, fern, palm frond, flowers, vine) whose alpha channel marks where the leaf is.

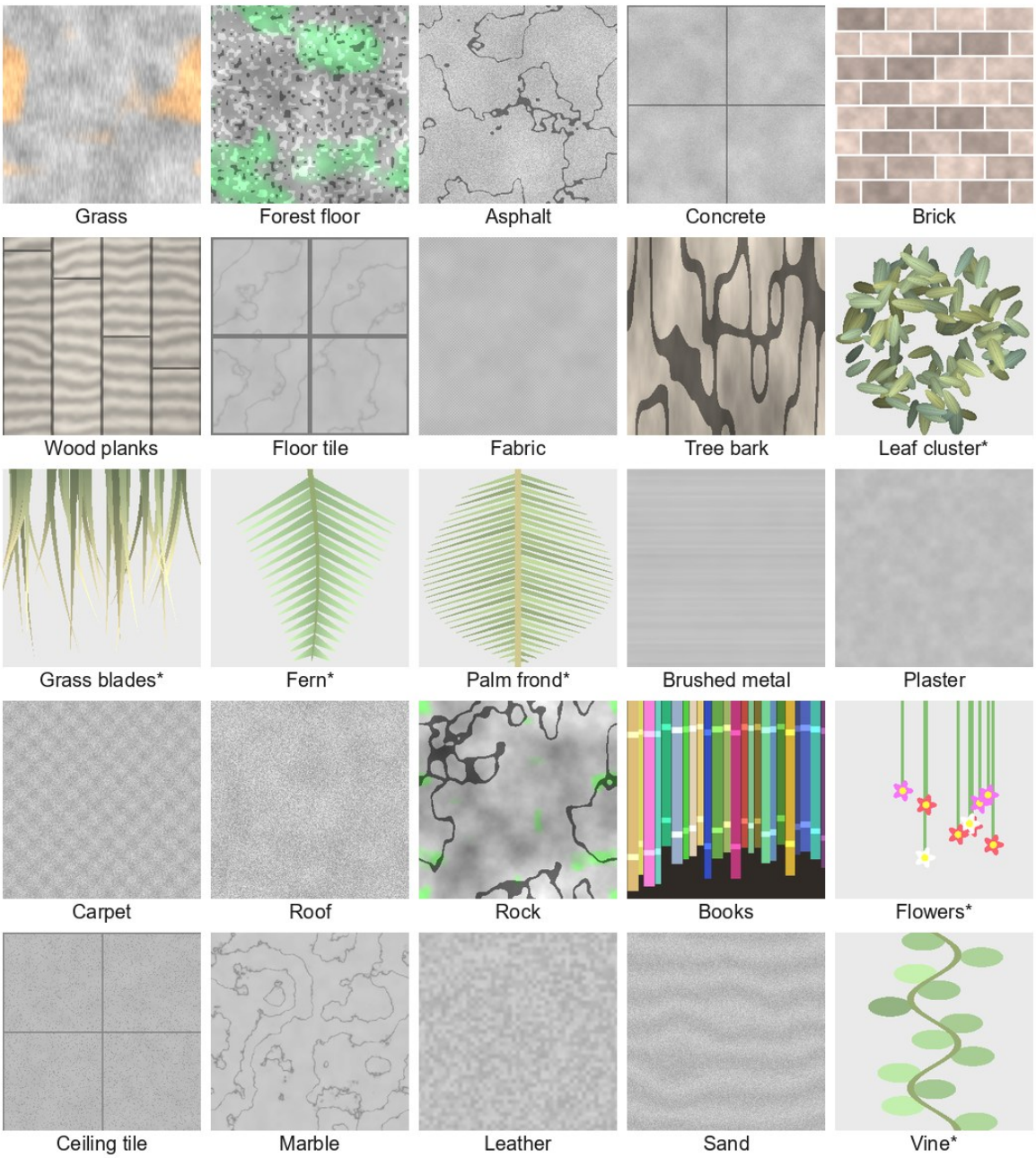


Figure 42: All procedural textures ($*$ = alpha cut-out, light grey shows the transparent part).

10.2 How textures are applied

- All layers live in one `GL_TEXTURE_2D_ARRAY` (256×256, mip-maps, $4\times$ anisotropic filtering). Each instance only stores a layer number, so a texture change never breaks an instanced draw call.
- A texture stores a *detail* value. The final colour is $k_d = \text{colour} \times 2 \cdot \text{texel}$, so the same brick texture works for any brick colour.
- **Object-space box mapping**: the shader picks the plane of the largest normal component and uses the local position multiplied by the object's scale: $(u, v) = (x, z)$, (z, y) or (x, y) times the repeats per metre. A 10 m wall therefore shows ten times more bricks than a 1 m box, without any UV work.
- Cut-out cards use their mesh UVs, and pixels with $\alpha < 0.5$ are discarded (also in the shadow pass, so leaves cast leaf-shaped shadows).

- The terrain mixes two textures (lawn grass and forest floor) using a large-scale noise and the distance from the clearing.

11 Shadows

Shadows use **shadow mapping** (Figure 43).

1. **Depth pass**: the scene is drawn from the sun with an orthographic camera (± 32 m box around the player, 2048×2048 depth texture, so one texel is about 3 cm). Only depth is stored.
2. **Main pass**: every pixel is moved into the light's space, $\mathbf{p}_L = P_L V_L \, (\mathbf{p} + 0.05\,\mathbf{N})$. If its depth is larger than the stored depth, it is in shadow.

To make the shadows look good:

- a slope-scaled bias $0.0006 + 0.0012(1 - \mathbf{N} \cdot \mathbf{L})$, a 5 cm normal offset and a polygon offset in the depth pass remove "shadow acne",
- **PCF**: 5 hardware-compared samples (the centre counts twice) give soft edges,
- no shimmering: the light box moves only in whole shadow-map texels (along the light's own axes), and the sun direction used for shadows changes only in steps of 0.5°, so shadow edges do not crawl,
- glass, grass and other tiny details do not cast shadows, which saves time.

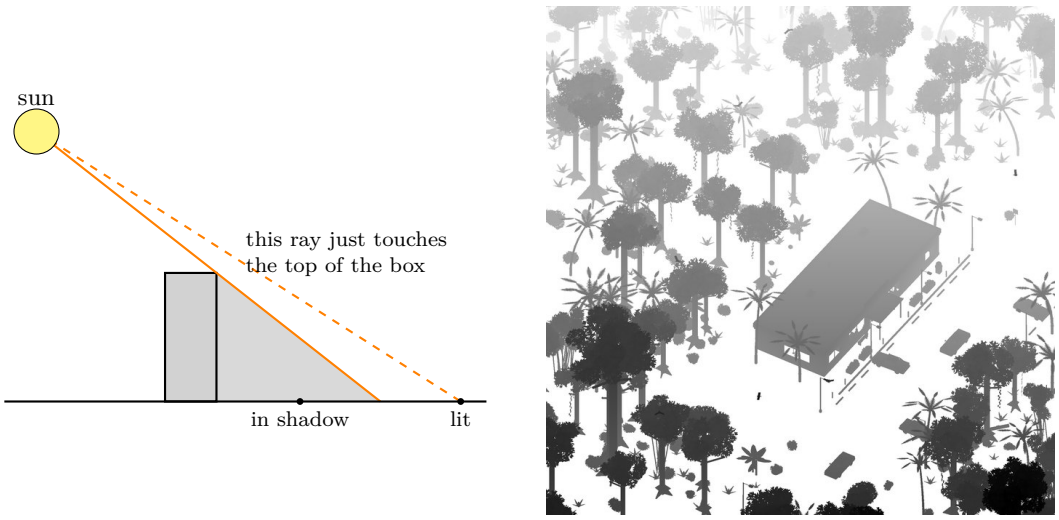


Figure 43: Left: the idea of shadow mapping. Right: the real depth image seen by the sun (darker = closer to the sun). The building, cars and every tree crown are visible.



(a) Shadows on **(b)** Shadows off

Figure 44: Shadows give the scene depth and show where objects touch the ground.

12 Sky, Fog, Tone Mapping and Wind

- **Sky**: one full-screen triangle. For each pixel the view ray is rebuilt from the inverse view-projection matrix. The colour is $\mathrm{mix}(\mathit{horizon}, \mathit{zenith}, h^{0.55})$, plus a sun disc $(\max(\mathbf{d} \cdot \mathbf{s}, 0)^{900})$ with a glow, a moon, twinkling stars (a hash on a 3D grid) and moving clouds (4-octave noise projected on a sky plane).
- **Fog**: $f = 1 - e^{-(d\rho)^{1.5}}$ blends to the horizon colour. It hides the end of the world and gives a humid jungle look. The density is higher at night and lower when the camera is high up.
- **Tone mapping**: $C' = 1 - e^{-C \cdot \mathit{exposure}}$ keeps bright lamps from turning flat white. The exposure rises at night, like an eye adapting to the dark.
- **Wind**: leaves and grass sway in the vertex shader. The offset grows with the height inside the shape: $\Delta x = 0.05\, h\, \min(s_y, 2.5) \sin(1.7t + 0.35x + 0.27z)$. Aquarium plants reuse this as a water current.

13 Basic Mode versus Enhanced Mode

Press **B** (or click the HUD button) to switch modes. **T**, **G** and **F** switch textures, shadows and fog one at a time.

Feature	Basic	Enhanced
Basic shapes, transformations, all objects	yes	yes
Blinn–Phong lighting with all lights	yes	yes
Procedural textures	–	yes
Shadow mapping (PCF)	–	yes
Fog	–	yes
Sky with sun, clouds, stars	simple gradient	yes
Wind animation	–	yes
Tone mapping	clamp	exponential



(a) Basic



(b) Enhanced



(c) Basic (interior)



(d) Enhanced without textures

Figure 45: Basic and Enhanced rendering.

14 Interaction: Using the Machines

14.1 Look-and-press picking

Every button, slot, flap, switch, TV or aquarium registers an *oriented box* (its matrix M and half size $\mathbf{h}$) every frame. A ray goes from the camera through the centre of the screen. For each box the ray is moved into the box's own space with M^{-1}, and the *slab test* is applied on each axis:

$$t_1 = \frac{-h_a - o_a}{d_a}, \quad t_2 = \frac{h_a - o_a}{d_a}, \qquad t_{near} = \max_a \min(t_1, t_2), \quad t_{far} = \min_a \max(t_1, t_2).$$

The ray hits the box if $t_{near} \le t_{far}$ and t_{near} is within reach (2.4 m). The closest box wins. Its prompt is shown under the crosshair (for example "Insert Tk 10 coin"), and it glows. Pressing **E** or the left mouse button runs its action. As an automatic test, the program aims the camera at fourteen different parts and records what it hits (Listing 1). Every part was hit correctly.

Listing 1: Automatic interaction test (part aimed at → prompt found)

```
vending coin slot -> [Insert Tk 10 coin (you have Tk 1120)]
vending note slot -> [Insert Tk 50 note (you have Tk 1120)]
vending key B -> [Press [B]]
vending key 2 -> [Press [2]]
vending OK -> [Press [OK]]
vending flap -> [Collection flap]
ATM card slot -> [Insert bank card]
ATM side key L1 -> [Side key L1]
ATM keypad 5 -> [Keypad [5]]
ATM cash slot -> [Cash dispenser]
coffee button 1 -> [Make Espresso (free)]
TV -> [Change TV channel (now: News)]
aquarium -> [Feed the fish]
light switch -> [Turn OFF lounge lights]
```

14.2 ATM: step by step

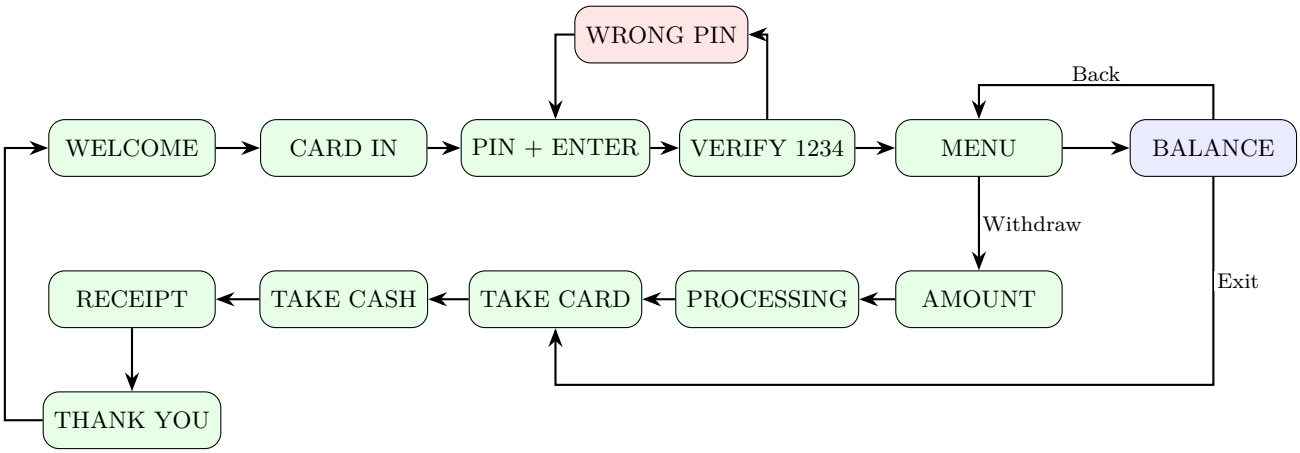


Figure 46: ATM state machine. CANCEL at any step jumps to TAKE CARD. The card is always returned before the cash, like a real ATM.

The same eight steps are listed in a HUD panel, and the current step is highlighted. Inside the machine, the card slides in and out, the PIN appears as stars, the bills slide out of the cash slot and the receipt paper curls out of the printer. Taking the cash adds Taka to the player's wallet and takes it from the bank balance. That money can then be spent at the vending machines.



(a) 1. Insert card



(b) 2. Enter PIN



(c) 3. Choose a service



(d) 4. Choose amount



(e) 5. Processing



(f) 6. Take card



(g) 7. Take cash



(h) 8. Take receipt



(i) Done

Figure 47: A full withdrawal of Tk 1,000, with the step list on the left.

14.3 Coffee machine and TV



(a) Brewing a cappuccino (stream, rising coffee, steam) **(b)** Ready, take the cup

Figure 48: Coffee machine: `IDLE` → `CUP_DROP` → `GRINDING` → `BREWING` → `READY`.



(a) News + ticker **(b)** Nature **(c)** Live football **(d)** Weather (game time)

Figure 49: The four animated TV channels.

15 NPC Visitors

15.1 Life of a visitor

Visitors can **only** appear inside a car at the far end of the road. A new car comes every 8–18 s if there is a free parking spot and the road entrance is clear. At most six visitors are active at once.

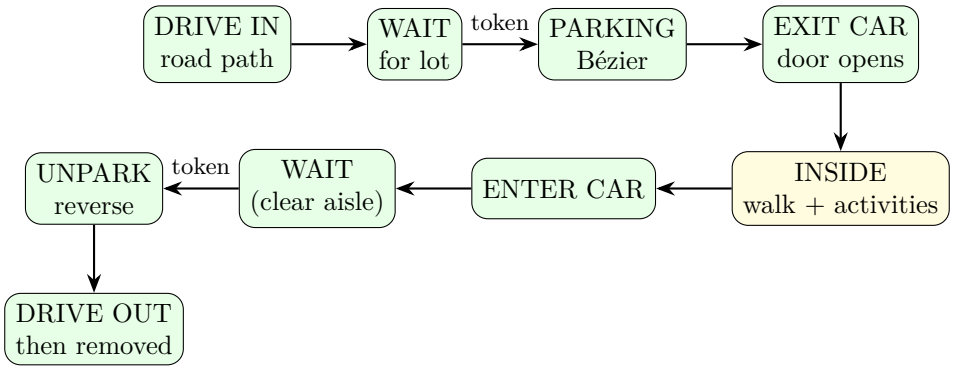


Figure 50: Visitor state machine.

15.2 Driving and parking

Cars follow *paths*: lists of points with the running length s. The car's position comes from interpolation at s, and its heading from the direction of the segment. The road path samples $z_{road}(x)$ every 3 m. Parking uses a cubic Bézier curve

$$\mathbf{B}(t) = (1-t)^3\mathbf{P}_0 + 3(1-t)^2 t\,\mathbf{P}_1 + 3(1-t)t^2\,\mathbf{P}_2 + t^3\mathbf{P}_3,$$

with $\mathbf{P}_0 = (x_s{+}5, 16)$ and $\mathbf{P}_1 = (x_s{+}2.2, 16)$ along the aisle, and $\mathbf{P}_2 = (x_s, z_s{\pm}2.8)$ and $\mathbf{P}_3 = (x_s, z_s)$ in the spot. The tangents at both ends therefore match the aisle and the spot, so the turn is smooth.

Leaving uses a second curve driven in *reverse* (the heading is the motion direction +180°, the reverse lights are on), and then the car drives forward to the west end of the road.

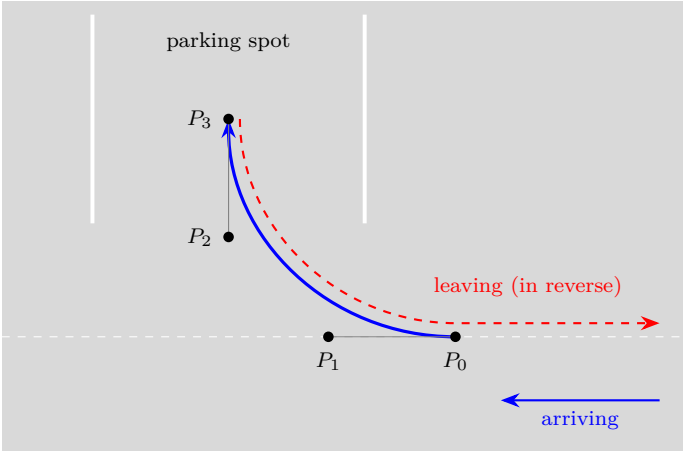


Figure 51: Parking (blue, forward) and leaving (red, reverse) curves for one spot.

Speed is limited by the place (9 m/s on the road, 4 m/s in the lot, 1.8 m/s while parking, 1.3 m/s while reversing) and by the distance s_{left} still to go: $v \le \sqrt{2 \cdot 2.2 \cdot s_{left}} + 0.25$, so cars brake smoothly. A car stops if another car is less than 7.5 m ahead in its lane, or if a person (NPC or player) is less than 5.5 m ahead. Only one car may park or unpark at a time: it must hold the "lot token". Cars in the aisle wait while the token holder moves in front of them, and a car only starts to reverse when the aisle behind it is clear. These two rules prevent deadlocks.

15.3 Walking and activities

Inside the building, people walk on a waypoint graph of 18 nodes and 22 edges (doorways, room hubs, the queue entrance at the ATM, and so on). The shortest route is found with **Dijkstra's algorithm**, and the route to the car is added at the end. Walking speed is 1.35 m/s, and people wait if the player blocks them. Every visitor has a plan of 1–3 activities. Each one *reserves* its resource first (a machine, a seat, a reading spot), so two people never use the same thing:

- **Vending**: walk to the machine, press the keys (press pose), wait for the drop, bend down, take the item and carry it. This really runs the machine's state machine.
- **Coffee**: choose a drink, wait, take the cup and carry it.
- **ATM**: run a complete ATM session. The ATM "presses its own buttons" with small pauses, so the screen shows every step.
- **Sit and watch TV** (7 sofa seats) or **read** (2 library armchairs) for 12–28 s.
- **Browse books** or **watch the fish**.



(a) Cars parked, a visitor walking to the entrance



(b) Walking in through the sliding doors



(c) Watching TV (the TV light flickers on them)



(d) Buying from a vending machine

Figure 52: NPC visitors.

Listing 2 shows the beginning of a real trace recorded by the program.

Listing 2: Start of a visitor trace (simulation time, visitor, spot, what it is doing)

```
t= 4.0s visitor 0 (spot 18): driving in
t= 8.7s visitor 1 (spot 10): driving in
t= 14.2s visitor 2 (spot 13): driving in
t= 19.7s visitor 0 (spot 18): waiting to park
t= 19.7s visitor 0 (spot 18): parking
t= 22.2s visitor 3 (spot 3): driving in
t= 24.5s visitor 0 (spot 18): getting out
t= 26.8s visitor 0 (spot 18): walking in
t= 27.3s visitor 4 (spot 0): driving in
t= 30.8s visitor 1 (spot 10): waiting to park
t= 30.8s visitor 1 (spot 10): parking
t= 32.9s visitor 5 (spot 14): driving in
t= 35.6s visitor 1 (spot 10): getting out
t= 37.9s visitor 1 (spot 10): walking in
t= 43.5s visitor 2 (spot 13): waiting to park
t= 43.5s visitor 2 (spot 13): parking
t= 47.5s visitor 0 (spot 18): browsing books
t= 48.3s visitor 2 (spot 13): getting out
t= 50.6s visitor 2 (spot 13): walking in
t= 51.3s visitor 3 (spot 3): waiting to park
t= 51.3s visitor 3 (spot 3): parking
t= 51.6s visitor 4 (spot 0): waiting to park
t= 54.1s visitor 1 (spot 10): watching fish
t= 56.1s visitor 3 (spot 3): getting out
t= 56.1s visitor 4 (spot 0): parking
t= 58.4s visitor 3 (spot 3): walking in
t= 60.9s visitor 4 (spot 0): getting out
t= 63.2s visitor 4 (spot 0): walking in
t= 69.6s visitor 5 (spot 14): waiting to park
t= 69.7s visitor 5 (spot 14): parking
t= 69.9s visitor 0 (spot 18): leaving
t= 73.4s visitor 2 (spot 13): at a vending machine
t= 73.9s visitor 1 (spot 10): leaving
t= 74.5s visitor 5 (spot 14): getting out
```

16 AI, Particles, Text and Other Effects

This section collects the "moving" parts of the program in one place, with the simple rule behind each.

16.1 All the AI in one table

None of it is complicated: every character is a small **state machine** (a list of states and the events that move it to the next state) plus a simple movement rule.

Table 39: Who moves by itself, and how

Who	Decides *what* to do	Decides *how* to move
Visitor (NPC)	state machine: drive in, park, get out, 1–3 activities, get in, leave (Figure 52); a resource (machine, seat) is reserved before use	shortest route on a waypoint graph with **Dijkstra**; walks at 1.35 m/s; waits if someone blocks the way
Car	part of the visitor's state machine; only the holder of the "lot token" may park or unpark	follows a path (road samples or a Bézier curve) by its length s; speed limit $v \le \sqrt{2a\,s_{left}}$ for smooth braking; stops behind cars (7.5 m) and people (5.5 m)
ATM / vending used by an NPC	the NPC calls the same functions the player's key presses call, with short pauses	–
Fish	swim on their own ellipse; if there is food, swim up to it	see below
Birds	fly in circles by day	see below
Fireflies	appear only at night, outdoors	see below

Fish. Each of the 7 fish has an angle φ that grows with its own speed, $\varphi \mathrel{+}= \omega\,\Delta t$ (ω is positive or negative, so some swim the other way). Its position is on an ellipse inside the tank, $x = r_x \cos\varphi,\ z = r_z \sin\varphi$, and it always faces the direction it moves: $yaw = \mathrm{atan2}(\Delta x, \Delta z)$. Its height moves slowly towards a target, $y_{target} = y_0 + 0.03\sin(0.7t + \varphi)$, at most 0.25 m/s. When you press "Feed the fish", 12 food crumbs appear at the surface and sink; while there is food, y_{target} becomes the water surface, so all fish swim up to eat. The tail is a child node that wags with $R_y(25^\circ \sin(9t + 3\varphi))$.

Birds. 9 birds fly on circles (radius 10–28 m, height 16–30 m) above the jungle: $\mathbf{p} = \mathbf{c} + (r\cos a,\ h + 1.5\sin 2a,\ r\sin a)$ with $a \mathrel{+}= \omega\Delta t$. A bird is a body sphere and a cone beak, and the two wings are cubes in child nodes rotated by $R_z(\pm 35^\circ \sin(\text{flap}))$, so they flap.

Fireflies. 110 small glowing spheres. Their positions are wrapped into a 60×60 m box around the camera ($x' = ((x - x_{cam}) \bmod 60) - 30 + x_{cam}$), so there are always some near the player, and they drift on sine curves. Each one blinks with $0.5 + 0.5\sin(3t + 3\varphi)$ and is hidden when this is below 0.25. They are only drawn at night and never inside the building.

16.2 Particles

A particle is a tiny basic shape with a position $\mathbf{p}$, a velocity $\mathbf{v}$ and a life time. Every frame: $\mathbf{v} \mathrel{+}= \mathbf{g}\,\Delta t$ (if it falls), $\mathbf{p} \mathrel{+}= \mathbf{v}\,\Delta t$, life $\mathrel{-}= \Delta t$; when the life is over it is removed.

Table 40: Particle effects

Effect	Shape	Rule
Coffee steam	see-through sphere	a new puff every 0.12 s while brewing; rises 7 cm/s and sways sideways; grows from 2 to 6 cm; alpha $0.25 \cdot$ life, so it fades out; lives 1.6 s
Change coins	thin cylinder	fall with gravity $g = 9.81$ m/s^2 into the coin cup and stop on its floor
Falling product	the product itself	gravity, spin, and a bounce on the tray: $v_y \leftarrow -0.32\, v_y$
Aquarium bubbles	see-through sphere	rise 12–22 cm/s with a small wobble; start again at the bottom when they reach the surface
Fish food	tiny sphere	sinks 4 cm/s, lives 5–8 s
Fireflies	glowing sphere	see above

16.3 Text rendering

All text (HUD, ATM and vending screens, TV, signs, price tags) uses one **font atlas**:

1. At start-up, Windows GDI draws the 95 printable ASCII characters (Consolas font) into one bitmap. The bitmap becomes an OpenGL texture whose alpha channel is the shape of the letters.

2. Each character on screen is one *textured quad* (two triangles) whose texture coordinates select that character's cell in the atlas. A string is a row of such quads.

3. **HUD text** uses an orthographic projection $\text{ortho}(0, w, h, 0)$, so one unit is one pixel; depth testing is off, so it is always on top.

4. **Text inside the 3D world** is drawn on a *canvas*: a flat rectangle with its own model matrix (for example the ATM screen). A canvas point (x, y) goes to the world with $\mathbf{p} = M_{canvas}\,(x - \frac{w}{2},\ y - \frac{h}{2},\ 0,\ 1)^T$, so the text sits exactly on the screen and turns with the machine. Depth testing is on, so a person walking in front hides the text.

Text is drawn with alpha blending, $C = \alpha\, C_{text} + (1 - \alpha)\, C_{behind}$, which gives smooth letter edges.

16.4 Collision detection

People (the player and NPCs) are **circles** on the ground. Walls, machines and furniture are **oriented boxes** with a height range, and tree trunks and lamp posts are circles. To push a person out of a box, the centre is turned into the box's own space (rotation by $-yaw$), the nearest point of the box is found by clamping, $\mathbf{q} = \text{clamp}(\mathbf{p}_{local}, -\mathbf{h}, \mathbf{h})$, and if $|\mathbf{p} - \mathbf{q}| < r$ the person is moved out along $\mathbf{p} - \mathbf{q}$. For two circles the rule is the same with the distance between the centres. This is repeated up to 3 times per frame, so corners work too. A box only counts if it overlaps the person's height (so you can walk under a table top), and the sliding doors switch their box off while they are open. The third-person camera casts a ray against the same boxes so it never goes through a wall.

16.5 Sound

Sound is kept small and simple. It uses 14 short **real recordings** that are free to use (CC0, public domain; from Kenney.nl and OpenGameArt, listed in `assets/sounds/CREDITS.txt`). They were trimmed, faded and levelled, and stored as 44.1 kHz mono WAV files that the program loads at start-up. The two background sounds (birds, crickets) are loops whose end is cross-faded into their start, so they repeat without a click. A small mixer thread adds all playing sounds together, limits the sum with tanh (no harsh clipping) and sends blocks of 2048 samples to the sound card with the Windows `waveOut` functions.

Table 41: Sounds

Sound	When
click, beep	any button or switch; ATM keys; coffee ready
coin, thud, motor	vending: money in, change falling, product hitting the tray, spiral turning
cash, grinder, pouring	ATM gives money; coffee machine grinds and pours
door whoosh, car door	entrance doors start to move; a visitor opens or closes the car door
birds / crickets	quiet background loop by day / by night, softer inside the building

Sounds in the world get quieter with distance d from the camera: $volume = v_0/(1 + (d/4)^2)$, so a machine used by a visitor in another room is only heard faintly. **O** switches sound off.

16.6 Fog, frame cap and timing

Fog is described in Section 12. All movement uses the real time step Δt of the frame, so the speed of everything is the same at 30 or 60 FPS. The frame cap measures how long the frame took, sleeps for most of the rest of the frame time, and then waits the last 1.5 ms in a short loop, so the frame rate is very steady (60 or 30 FPS, selectable).

17 The Player and the Cameras

- **First person**: the camera is at eye height (1.62 m), with mouse look and a small head bob $0.028 \sin(2\varphi)$ while walking.
- **Third person**: the player's own model is visible. The camera sits 3.4 m behind the head. A ray from the head to the camera is tested against the walls, and the camera moves closer if a wall is in the way.
- **Free camera**: fly anywhere (W A S D, Space and Ctrl) to inspect the scene, for example with the roof cut-away view.
- **FOV**: the mouse wheel or $+/-$ changes the vertical field of view from 25° to 110°. This changes the $1/\tan(f/2)$ terms of the projection matrix.

Collision. The player is a circle with a radius of 0.3 m. For every obstacle box (walls, furniture, machines, cars), the circle centre is moved into the box's space, the closest point $\mathbf{q} = \text{clamp}(\mathbf{p}, -\mathbf{h}, \mathbf{h})$ is found, and if $|\mathbf{p} - \mathbf{q}| < r$ the circle is pushed out along $\mathbf{p} - \mathbf{q}$ (three passes). Tree trunks and people are circles. Gravity and jumping use $v_y \mathrel{-}= 12\,\Delta t$ on top of the terrain height.



(a) Third person



(b) Free camera, roof cut-away, night



(c) FOV 35°



(d) FOV 70°



(e) FOV 100°

Figure 53: Camera modes and field of view.

18 Performance on a Weak GPU

18.1 Techniques

- **Instanced drawing**: all copies of one basic shape are drawn with one `glDrawElementsInstanced` call. Each instance carries 37 floats (model matrix 16, normal matrix 9, colour and alpha 4, emission and texture layer 4, specular, shininess, texture scale and flags 4).
- **Static world in chunks**: 84,918 shapes that never move are collected once (in 0.66 s) into 101 chunks of 32×32 m, each with its own GPU buffer.
- **Culling**: a chunk is skipped if its bounding box is outside the view frustum (6 planes taken from the view-projection matrix) or more than 165 m away. Small things have shorter ranges: grass and flowers 45 m, and bushes, ferns and leaf cards 85 m.
- **Front-to-back order + occlusion culling**: visible chunks are drawn from near to far. For every chunk the GPU answers an *occlusion query* ("was any pixel of it visible?"). A chunk that was completely hidden last frame (for example the jungle behind the back wall when you stand inside) is skipped. Only its bounding box is drawn, invisibly, to ask again. Results are read one frame later, so the CPU never waits for the GPU.
- **Depth pre-pass**: the scene is first drawn with a tiny depth-only shader. Then the full lighting shader runs with `GL_LEQUAL` and depth writing off, so *each pixel is lit exactly once*, instead of lighting walls, floors and leaves that are covered later. Both shaders mark `gl_Position` as `invariant`, so the depths match exactly.
- **Shadow pass** only draws chunks inside the light's box, and skips grass and glass.
- **Per-room light lists** and frustum culling of lights (see the Lighting section).
- **Moving objects** are written into streaming buffers once per frame. Transparent objects are sorted from back to front.
- **Frame cap**: after each frame the program sleeps until $1/cap - 1.5$ ms and then waits the last moment in a loop (with a 1 ms Windows timer resolution). This gives a very steady 60 or 30

FPS and keeps the laptop cool.

- **Low mode** (`-low`): a 1024 shadow map, fewer plants, fewer visitors and no MSAA.

18.2 Measured speed

Measured by the program itself without the frame cap (1280×720, Intel Iris Xe). Every view stays above 60 FPS, so the 60 FPS cap always holds.

Table 42: Frame time without the cap

View	Enhanced		Basic		Triangles
	ms	FPS	ms	FPS	(enh.)
Outside, day	13.9	72	11.5	87	2.27M
Main lounge	12.2	82	9.3	108	2.23M
Inside the jungle	10.8	93	8.1	124	1.80M
Parking lot, night	11.5	87	11.1	90	1.70M
Aerial view	16.0	63	13.2	76	2.17M

19 Conclusion

Jungle Rest Stop is a small living world: three connected rooms with working machines, visitors who come by car and act on their own, a dense jungle and light that changes from morning to night. Every object is a set of basic shapes placed with translation, rotation and scaling, organised in parent and child nodes. The project covers the main topics of a graphics course in one program: transformations, hierarchical modelling, illumination and shading, light types, textures, shadow mapping, fog, transparency, animation, picking, collision and cameras, and it runs at a steady 60 FPS on integrated graphics.

References

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